

Predator prey model with distributed delay

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We consider the model

$$\begin{cases} \frac{dh_i}{dt} = \frac{h_i}{\varepsilon} \left((1 - \alpha h_i) - \frac{p_i}{1 + h_i} \right) + d \left(\int_0^\infty h_j(t-s) g_a^p(s) ds - h_i(t) \right), & i, j = 1, 2, \ i \neq j, \\ \frac{dp_i}{dt} = p_i \left(\frac{h_i}{1 + h_i} - \mu \right), \end{cases} \quad (1)$$

where $g_a^p(s) := \frac{a^p s^{p-1} e^{-as}}{\Gamma(p)}$ is the kernel of the Gamma distribution with $\Gamma(p) = (p-1)!$ for integer values of p and $\int_0^\infty g_a^p(s) ds = 1$.

1 Existence, positivity and boundedness of solutions

1.1 Existence of solution

In this section, we show that solution of (1) exists. To do so, we define

$$X = \begin{pmatrix} h_1 \\ p_1 \\ h_2 \\ p_2 \end{pmatrix} \quad \text{and} \quad F(X) = \begin{pmatrix} \frac{h_1}{\varepsilon} \left((1 - \alpha h_1) - \frac{p_1}{1 + h_1} \right) + d \left(\int_0^\infty h_2(t-s) g_a^p(s) ds - h_1(t) \right) \\ p_1 \left(\frac{h_1}{1 + h_1} - \mu \right) \\ \frac{h_2}{\varepsilon} \left((1 - \alpha h_2) - \frac{p_2}{1 + h_2} \right) + d \left(\int_0^\infty h_1(t-s) g_a^p(s) ds - h_2(t) \right) \\ p_2 \left(\frac{h_2}{1 + h_2} - \mu \right) \end{pmatrix} := \begin{pmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{pmatrix}$$

and write (1) as $\frac{dX}{dt} = F(X)$. We consider the Banach space $B := \mathcal{C}((-\infty, 0], \mathbb{R}^4)$ endowed with the following norm $\|\phi\| = \sup_{\theta \in (-\infty, 0]} |\phi(\theta)|$. We recall that $|\cdot|$ is any norm on $\mathbb{R}^4$ or the euclidean norm. We define $B_+ := \{\phi \in B : \phi_i(\theta) \geq 0, i = 1, \dots, 4, \theta \leq 0\}$.

Next, we will show that $F_i, i = 1, \dots, 4$, satisfies the Lipschitz condition on each variable.

1.2 Positivity of solutions

Let's assume that $h_i(0) > 0$ and $p_i(0) > 0, i = 1, 2$.

First, we show by contradiction that $h_i(t) > 0, i = 1, 2$. Suppose that $h_i(t)$ is not positive. Hence, there exists $t_i > 0$ such that $h_i(t) > 0$ when $t \in [0, t_i - \delta]$, $h_i(t_i) = 0$ and $h_i(t_i) < 0$ when $t \in (t_i, t_i + \delta]$. Then $\frac{dh_i}{dt}(t) \leq 0$ for $t \in [0, t_i]$. Without loss of generality, we assume $t_1 \leq t_2$.

Substituting $h_i(t_i)$ in the first equation of (1) gives $\frac{dh_1}{dt}(t_1) = d \int_0^\infty h_2(t_1-s) g_a^p(s) ds$ and $\frac{dh_2}{dt}(t_2) = d \int_0^\infty h_1(t_2-s) g_a^p(s) ds$.

Since $t_1 - s \leq t_2$ and $t_2 - s \leq t_2$, we have $h_2(t_1-s) > 0$ and $h_2(t_s-s) > 0$. This implies that $\int_0^\infty h_2(t_1-s) g_a^p(s) ds > 0$ and $\int_0^\infty h_2(t_2-s) g_a^p(s) ds > 0$. Therefore, $\frac{dh_i}{dt}(t_i) > 0, i = 1, 2$. This is a contradiction. We then conclude that h_i is positive.

Second, we show that $p_i(t) > 0, i = 1, 2$. The solution of the second equation of (1) is $p_i(t) = p_i(0) e^{-\mu t} \exp \left(\int_0^t \frac{h_i(s)}{1 + h_i(s)} ds \right)$, which is positive since $p_i(0) > 0$ and $h_i > 0, i = 1, 2$.

We then conclude that solutions of (1) are all positive.

1.3 Boundedness of solutions

Here, we will show that the functions h_i and p_i , $i = 1, 2$, are bounded by showing that their sum is.

We define $N(t) = h_1(t) + \frac{1}{\varepsilon}p_1(t) + h_2(t) + \frac{1}{\varepsilon}p_2(t)$. Then,

$$\begin{aligned} \frac{dN}{dt} &= \frac{dh_1}{dt} + \frac{1}{\varepsilon} \frac{dp_1}{dt} + \frac{dh_2}{dt} + \frac{1}{\varepsilon} \frac{dp_2}{dt} \\ &= \frac{h_1}{\varepsilon} - \alpha \frac{h_1^2}{\varepsilon} - \frac{p_1}{\varepsilon} \frac{h_1}{1+h_1} + d \int_0^\infty h_2(t-s) g_a^p(s) ds - dh_1(t) + \frac{p_1}{\varepsilon} \frac{h_1}{1+h_1} - \mu \frac{p_1}{\varepsilon} \\ &\quad + \frac{h_2}{\varepsilon} - \alpha \frac{h_2^2}{\varepsilon} - \frac{p_2}{\varepsilon} \frac{h_2}{1+h_2} + d \int_0^\infty h_1(t-s) g_a^p(s) ds - dh_2(t) + \frac{p_2}{\varepsilon} \frac{h_2}{1+h_2} - \mu \frac{p_2}{\varepsilon} \\ &= \frac{h_1}{\varepsilon} - \alpha \frac{h_1^2}{\varepsilon} + d \int_0^\infty h_2(t-s) g_a^p(s) ds - dh_1(t) - \mu \frac{p_1}{\varepsilon} + \frac{h_2}{\varepsilon} - \alpha \frac{h_2^2}{\varepsilon} + d \int_0^\infty h_1(t-s) g_a^p(s) ds - dh_2(t) - \mu \frac{p_2}{\varepsilon} \\ \frac{dN}{dt} &\leq \frac{h_1}{\varepsilon} - \alpha \frac{h_1^2}{\varepsilon} + dH(t) \int_0^\infty g_a^p(s) ds - dh_1(t) - \mu \frac{p_1}{\varepsilon} + \frac{h_2}{\varepsilon} - \alpha \frac{h_2^2}{\varepsilon} + dH(t) \int_0^\infty g_a^p(s) ds - dh_2(t) - \mu \frac{p_2}{\varepsilon} \end{aligned}$$

where $H(t) = \max_{s \leq t} (h_1(s), h_2(s))$. Since $\int_0^\infty g_a^p(s) ds = 1$, we then obtain:

$$\begin{aligned} \frac{dN}{dt} &\leq \frac{h_1}{\varepsilon} - \alpha \frac{h_1^2}{\varepsilon} - dh_1(t) - \mu \left(\frac{p_1}{\varepsilon} + \frac{p_2}{\varepsilon} \right) + \frac{h_2}{\varepsilon} - \alpha \frac{h_2^2}{\varepsilon} - dh_2(t) + 2dH(t) \\ &\leq \frac{h_1}{\varepsilon} - \alpha \frac{h_1^2}{\varepsilon} - dh_1(t) - \mu (N(t) - h_1(t) - h_2(t)) + \frac{h_2}{\varepsilon} - \alpha \frac{h_2^2}{\varepsilon} - dh_2(t) + 2dH(t), \quad \text{using the definition of } N(t) \\ &\leq -\mu N(t) + \frac{h_1}{\varepsilon} - \alpha \frac{h_1^2}{\varepsilon} + \mu h_1(t) - dh_1(t) + \frac{h_2}{\varepsilon} - \alpha \frac{h_2^2}{\varepsilon} + \mu h_2(t) - dh_2(t) + 2dH(t) \\ &\leq -\mu N + \frac{\alpha}{\varepsilon} h_1 \left(\frac{1 + \varepsilon(\mu - d)}{\alpha} - h_1 \right) + \frac{\alpha}{\varepsilon} h_2 \left(\frac{1 + \varepsilon(\mu - d)}{\alpha} - h_2 \right) + 2dH(t) \end{aligned}$$

Since $0 \leq \left(\frac{1 + \varepsilon(\mu - d)}{2\alpha} - h_1 \right)^2 = \left(\frac{1 + \varepsilon(\mu - d)}{2\alpha} \right)^2 + h_1^2 - h_1 \left(\frac{1 + \varepsilon(\mu - d)}{\alpha} \right)$, we have $\left(\frac{1 + \varepsilon(\mu - d)}{2\alpha} \right)^2 \geq h_1 \left(\frac{1 + \varepsilon(\mu - d)}{\alpha} - h_1 \right)$.

Similarly, we obtain $\left(\frac{1 + \varepsilon(\mu - d)}{2\alpha} \right)^2 \geq h_2 \left(\frac{1 + \varepsilon(\mu - d)}{\alpha} - h_2 \right)$. Hence,

$$\frac{dN}{dt} \leq -\mu N + 2 \frac{\alpha}{\varepsilon} \left(\frac{1 + \varepsilon(\mu - d)}{2\alpha} \right)^2 + 2dH(t) \leq -\mu N + 2 \left(\frac{\alpha}{\varepsilon} \left(\frac{1 + \varepsilon(\mu - d)}{2\alpha} \right)^2 + d \max \left(H(0), \frac{1}{\alpha} \right) \right), \quad (2)$$

since $H(t) \leq \max \left(H(0), \frac{1}{\alpha} \right)$.

From (2), we obtain that

$$\begin{aligned}
(2) &\implies \frac{dN}{dt} + \mu N \leq 2 \max \left(\frac{\alpha}{\varepsilon} \left(\frac{1 + \varepsilon(\mu - d)}{2\alpha} \right)^2 + dH(0), \frac{\alpha}{\varepsilon} \left(\frac{1 + \varepsilon(\mu - d)}{2\alpha} \right)^2 + d\frac{1}{\alpha} \right) \\
&\implies e^{\mu t} \left(\frac{dN}{dt} + \mu N \right) \leq 2e^{\mu t} \max \left(\frac{\alpha}{\varepsilon} \left(\frac{1 + \varepsilon(\mu - d)}{2\alpha} \right)^2 + dH(0), \frac{\alpha}{\varepsilon} \left(\frac{1 + \varepsilon(\mu - d)}{2\alpha} \right)^2 + d\frac{1}{\alpha} \right) \\
&\implies \frac{d(e^{\mu t} N)}{dt} \leq 2e^{\mu t} \max \left(\frac{\alpha}{\varepsilon} \left(\frac{1 + \varepsilon(\mu - d)}{2\alpha} \right)^2 + dH(0), \frac{\alpha}{\varepsilon} \left(\frac{1 + \varepsilon(\mu - d)}{2\alpha} \right)^2 + d\frac{1}{\alpha} \right) \\
&\implies \int_0^t \frac{d(e^{\mu s} N)}{ds} ds \leq 2 \max \left(\frac{\alpha}{\varepsilon} \left(\frac{1 + \varepsilon(\mu - d)}{2\alpha} \right)^2 + dH(0), \frac{\alpha}{\varepsilon} \left(\frac{1 + \varepsilon(\mu - d)}{2\alpha} \right)^2 + d\frac{1}{\alpha} \right) \int_0^t e^{\mu s} ds \\
&\implies e^{\mu t} N(t) - N(0) \leq \frac{2}{\mu} (e^{\mu t} - 1) \max \left(\frac{\alpha}{\varepsilon} \left(\frac{1 + \varepsilon(\mu - d)}{2\alpha} \right)^2 + dH(0), \frac{\alpha}{\varepsilon} \left(\frac{1 + \varepsilon(\mu - d)}{2\alpha} \right)^2 + d\frac{1}{\alpha} \right) \\
&\implies N(t) \leq N(0)e^{-\mu t} + \frac{2}{\mu} (1 - e^{-\mu t}) \max \left(\frac{\alpha}{\varepsilon} \left(\frac{1 + \varepsilon(\mu - d)}{2\alpha} \right)^2 + dH(0), \frac{\alpha}{\varepsilon} \left(\frac{1 + \varepsilon(\mu - d)}{2\alpha} \right)^2 + d\frac{1}{\alpha} \right)
\end{aligned}$$

We then conclude that $\limsup_{t \rightarrow \infty} N(t) \leq \frac{2}{\mu} \max \left(\frac{\alpha}{\varepsilon} \left(\frac{1 + \varepsilon(\mu - d)}{2\alpha} \right)^2 + dH(0), \frac{\alpha}{\varepsilon} \left(\frac{1 + \varepsilon(\mu - d)}{2\alpha} \right)^2 + d\frac{1}{\alpha} \right)$.

2 Steady-state solutions of (1)

Let $(\bar{h}_1, \bar{p}_1, \bar{h}_2, \bar{p}_2)$ be the equilibrium point of (1). It satisfies the following

$$\begin{cases} \frac{\bar{h}_i}{\varepsilon} \left((1 - \alpha \bar{h}_i) - \frac{\bar{p}_i}{1 + \bar{h}_i} \right) + d(\bar{h}_j - \bar{h}_i) = 0, \\ \bar{p}_i \left(\frac{\bar{h}_i}{1 + \bar{h}_i} - \mu \right) = 0. \end{cases} \quad (3a)$$

Solving equation (3b) gives $\bar{p}_i = 0$. It also gives $\bar{h}_i = \frac{\mu}{1 - \mu}$ for which we require $\mu < 1$ since we are looking for positive solutions. Next, we solve (3a) for each of these values.

1. Putting $\bar{p}_i = 0$, $i = 1, 2$, into equation (3a) gives $\bar{h}_i(1 - \alpha \bar{h}_i) + \varepsilon d(\bar{h}_j - \bar{h}_i) = 0$. For $i = 1, 2$, we obtain

$$\begin{cases} \alpha \bar{h}_1^2 + (\varepsilon d - 1)\bar{h}_1 - \varepsilon d\bar{h}_2 = 0 \\ \alpha \bar{h}_2^2 + (\varepsilon d - 1)\bar{h}_2 - \varepsilon d\bar{h}_1 = 0. \end{cases} \quad (4a)$$

$$\quad (4b)$$

Their subtraction gives

$$\begin{aligned}
&\alpha(\bar{h}_1^2 - \bar{h}_2^2) + (\varepsilon d - 1)(\bar{h}_1 - \bar{h}_2) - \varepsilon d(\bar{h}_2 - \bar{h}_1) = 0 \\
&\alpha(\bar{h}_1 - \bar{h}_2)(\bar{h}_1 + \bar{h}_2) + (\varepsilon d - 1)(\bar{h}_1 - \bar{h}_2) + \varepsilon d(\bar{h}_1 - \bar{h}_2) = 0 \\
&(\bar{h}_1 - \bar{h}_2) [\alpha(\bar{h}_1 + \bar{h}_2) + (2\varepsilon d - 1)] = 0 \\
&\bar{h}_1 = \bar{h}_2 \quad \text{or} \quad \bar{h}_1 + \bar{h}_2 = \frac{1 - 2\varepsilon d}{\alpha}.
\end{aligned}$$

- $\bar{h}_1 = \bar{h}_2$ in (4a) gives $\alpha \bar{h}_1^2 - \bar{h}_1 = 0 \implies \bar{h}_1 (\alpha \bar{h}_1 - 1) = 0 \implies \bar{h}_1 = 0$ or $\bar{h}_1 = \frac{1}{\alpha}$.
- In the condition $\bar{h}_1 + \bar{h}_2 = \frac{1 - 2\varepsilon d}{\alpha}$, if $\varepsilon d = \frac{1}{2}$, then $\bar{h}_1 + \bar{h}_2 = 0$. This implies that $\bar{h}_1 = -\bar{h}_2$. Therefore, we cannot have $\bar{h}_1$ and $\bar{h}_2$ both positive. Since we are looking for positive solutions, we require $\varepsilon d < \frac{1}{2}$. Let's assume that

$\varepsilon d < \frac{1}{2}$. We have $\bar{h}_1 = \frac{1-2\varepsilon d}{\alpha} - \bar{h}_2$ and

$$\begin{aligned} (4b) &\implies \alpha \bar{h}_2^2 + (\varepsilon d - 1) \bar{h}_2 - \varepsilon d \left(\frac{1-2\varepsilon d}{\alpha} - \bar{h}_2 \right) = 0 \\ &\implies \alpha^2 \bar{h}_2^2 + \alpha(\varepsilon d - 1) \bar{h}_2 - \varepsilon d (1 - 2\varepsilon d - \alpha \bar{h}_2) = 0 \\ &\implies \alpha^2 \bar{h}_2^2 + \alpha(2\varepsilon d - 1) \bar{h}_2 + \varepsilon d (2\varepsilon d - 1) = 0 \end{aligned}$$

The discriminant of this latter equation is given by

$$\Delta = \alpha^2 (2\varepsilon d - 1)^2 - 4\alpha^2 \varepsilon d (2\varepsilon d - 1) = \alpha^2 (2\varepsilon d - 1) (2\varepsilon d - 1 - 4\varepsilon d) = \alpha^2 (1 - 2\varepsilon d) (1 + 2\varepsilon d) = \alpha^2 (1 - 4\varepsilon^2 d^2) > 0.$$

Since the discriminant is positive, we obtain two real roots as:

$$\bar{h}_2^\pm = \frac{\alpha(1 - 2\varepsilon d) \pm \alpha\sqrt{1 - 4\varepsilon^2 d^2}}{2\alpha^2} = \frac{(1 - 2\varepsilon d) \pm \sqrt{1 - 4\varepsilon^2 d^2}}{2\alpha}.$$

It is easy to show for $\varepsilon d < \frac{1}{2}$ that $\sqrt{1 - 4\varepsilon^2 d^2} > 1 - 2\varepsilon d > 0$. Hence, $\bar{h}_2^- < 0 < \bar{h}_2^+$. We only consider the expression of $\bar{h}_2^+$, but with this, we find that

$$\bar{h}_1 = \frac{1 - 2\varepsilon d}{\alpha} - \frac{(1 - 2\varepsilon d) + \sqrt{1 - 4(\varepsilon d)^2}}{2\alpha} = \frac{2 - 4\varepsilon d - 1 + 2\varepsilon d - \sqrt{1 - 4(\varepsilon d)^2}}{2\alpha} = \bar{h}_2^- < 0.$$

Thus if $\varepsilon d < \frac{1}{2}$ then when $\bar{h}_2$ is positive, $\bar{h}_1$ is negative. We can't have $\bar{h}_1$ and $\bar{h}_2$ both positive such that $\bar{h}_1 + \bar{h}_2 = \frac{1 - 2\varepsilon d}{\alpha}$ is positive.

In conclusion, we find the following equilibria $(0, 0, 0, 0)$ and $\left(\frac{1}{\alpha}, 0, \frac{1}{\alpha}, 0\right)$.

2. Putting $\bar{h}_i = \frac{\mu}{1 - \mu}$, $i = 1, 2$, $\mu < 1$, in (3a) gives

$$\bar{h}_i(1 - \alpha \bar{h}_i) - \frac{\bar{p}_i \bar{h}_i}{1 + \bar{h}_i} = 0 \implies \bar{p}_i = \frac{\bar{h}_i(1 - \alpha \bar{h}_i)}{\frac{\bar{h}_i}{1 + \bar{h}_i}} = \frac{\frac{\mu}{1 - \mu} \left(1 - \alpha \frac{\mu}{1 - \mu}\right)}{\mu} = \frac{1 - \mu(1 + \alpha)}{(1 - \mu)^2}.$$

We therefore obtain the equilibrium point $\left(\frac{\mu}{1 - \mu}, \frac{\mu}{1 - \mu}, \frac{1 - \mu(1 + \alpha)}{(1 - \mu)^2}, \frac{1 - \mu(1 + \alpha)}{(1 - \mu)^2}\right)$, for which we require $\mu < \frac{1}{1 + \alpha}$.

3. We consider the case when $\bar{p}_i = 0$ and $\bar{h}_j = \frac{\mu}{1 - \mu}$, $i, j = 1, 2$, $i \neq j$, $\mu < 1$. Equation (3a) reads

$$\frac{1}{\varepsilon} \bar{h}_i(1 - \alpha \bar{h}_i) + d(\bar{h}_j - \bar{h}_i) = 0 \implies \bar{h}_i - \alpha \bar{h}_i^2 + \varepsilon d \bar{h}_j - \varepsilon d \bar{h}_i = 0 \implies \alpha \bar{h}_i^2 + (\varepsilon d - 1) \bar{h}_i - \varepsilon d \bar{h}_j = 0.$$

The discriminant is $\Delta = (\varepsilon d - 1)^2 + 4\alpha \varepsilon d \bar{h}_j$ and the positive root is given by $\bar{h}_i = \frac{1 - \varepsilon d + \sqrt{(\varepsilon d - 1)^2 + 4\alpha \varepsilon d \bar{h}_j}}{2\alpha}$.

Next, we find $\bar{p}_j$ by solving $\frac{1}{\varepsilon} \left(\bar{h}_j(1 - \alpha\bar{h}_j) - \frac{\bar{p}_j\bar{h}_j}{1 + \bar{h}_j} \right) + d(\bar{h}_i - \bar{h}_j) = 0$.

$$\begin{aligned}
&\implies \bar{h}_j(1 - \alpha\bar{h}_j) - \frac{\bar{p}_j\bar{h}_j}{1 + \bar{h}_j} + \varepsilon d(\bar{h}_i - \bar{h}_j) = 0 \\
&\implies \frac{\bar{p}_j\bar{h}_j}{1 + \bar{h}_j} = \bar{h}_j(1 - \alpha\bar{h}_j) + \varepsilon d(\bar{h}_i - \bar{h}_j) \\
&\implies \bar{p}_j = \frac{1 + \bar{h}_j}{\bar{h}_j} [\bar{h}_j(1 - \alpha\bar{h}_j - \varepsilon d) + \varepsilon d\bar{h}_i] \\
&\implies \bar{p}_j = \frac{1}{\mu} (-\alpha\bar{h}_j^2 + \bar{h}_j(1 - \varepsilon d) + \varepsilon d\bar{h}_i) \\
&\implies \bar{p}_j = -\frac{1}{\mu} (\alpha\bar{h}_j^2 - \bar{h}_j(1 - \varepsilon d) - \varepsilon d\bar{h}_i)
\end{aligned}$$

In conclusion, we have the following two equilibria:

$$(\bar{h}_1, \bar{p}_1, \bar{h}_2, 0) \text{ with } \bar{h}_1 = \frac{\mu}{1 - \mu}, \bar{h}_2 = \frac{1 - \varepsilon d + \sqrt{(\varepsilon d - 1)^2 + 4\alpha\varepsilon\bar{h}_1}}{2\alpha}, \bar{p}_1 = -\frac{1}{\mu} (\alpha\bar{h}_1^2 - \bar{h}_1(1 - \varepsilon d) - \varepsilon d\bar{h}_2)$$

and

$$(\bar{h}_1, 0, \bar{h}_2, \bar{p}_2) \text{ with } \bar{h}_2 = \frac{\mu}{1 - \mu}, \bar{h}_1 = \frac{1 - \varepsilon d + \sqrt{(\varepsilon d - 1)^2 + 4\alpha\varepsilon\bar{h}_2}}{2\alpha}, \bar{p}_2 = -\frac{1}{\mu} (\alpha\bar{h}_2^2 - \bar{h}_2(1 - \varepsilon d) - \varepsilon d\bar{h}_1).$$

3 Linear stability of steady-state solutions

3.1 The eigenvalue problem

We linearize system (1) at its equilibrium point $(\bar{h}_1, \bar{p}_1, \bar{h}_2, \bar{p}_2)$ by setting $h_i(t) = \bar{h}_i + \xi_i e^{\lambda t}$ and $p_i(t) = \bar{p}_i + \theta_i e^{\lambda t}$, $i = 1, 2$.

The first equation of (1) becomes

$$\begin{aligned}
\frac{d(\bar{h}_i + \xi_i e^{\lambda t})}{dt} &= \frac{1}{\varepsilon} \left((\bar{h}_i + \xi_i e^{\lambda t}) (1 - \alpha (\bar{h}_i + \xi_i e^{\lambda t})) - \frac{(\bar{p}_i + \theta_i e^{\lambda t}) (\bar{h}_i + \xi_i e^{\lambda t})}{1 + (\bar{h}_i + \xi_i e^{\lambda t})} \right) \\
&\quad + d \left(\int_0^\infty (\bar{h}_j + \xi_j e^{\lambda(t-s)}) g_a^p(s) ds - (\bar{h}_i + \xi_i e^{\lambda t}) \right) \\
\lambda \xi_i e^{\lambda t} &= \frac{1}{\varepsilon} \left((\bar{h}_i (1 - \alpha \bar{h}_i) - \alpha \bar{h}_i \xi_i e^{\lambda t} + \xi_i e^{\lambda t} - \alpha \bar{h}_i \xi_i e^{\lambda t}) - \frac{\bar{p}_i \bar{h}_i + \bar{p}_i \xi_i e^{\lambda t} + \bar{h}_i \theta_i e^{\lambda t}}{1 + \bar{h}_i + \xi_i e^{\lambda t}} + \mathcal{O}(\xi_i \theta_i + \xi^2) \right) \\
&\quad + d \bar{h}_j + d \xi_j \int_0^\infty e^{\lambda(t-s)} g_a^p(s) ds - d \bar{h}_i - d \xi_i e^{\lambda t} \\
&= \left(\frac{1}{\varepsilon} (\bar{h}_i (1 - \alpha \bar{h}_i)) + d \bar{h}_j - d \bar{h}_i \right) + \frac{1}{\varepsilon} \left((-2\alpha \bar{h}_i \xi_i e^{\lambda t} + \xi_i e^{\lambda t}) - \frac{\bar{p}_i \bar{h}_i + \bar{p}_i \xi_i e^{\lambda t} + \bar{h}_i \theta_i e^{\lambda t}}{1 + \bar{h}_i + \xi_i e^{\lambda t}} + \mathcal{O}(\xi_i \theta_i + \xi^2) \right) \\
&\quad + d \xi_j \int_0^\infty e^{\lambda t} e^{-\lambda s} g_a^p(s) ds - d \xi_i e^{\lambda t} \\
&= \left(\frac{1}{\varepsilon} \frac{\bar{p}_i \bar{h}_i}{1 + \bar{h}_i} \right) + \frac{1}{\varepsilon} \left((-2\alpha \bar{h}_i \xi_i e^{\lambda t} + \xi_i e^{\lambda t}) - \frac{\bar{p}_i \bar{h}_i + \bar{p}_i \xi_i e^{\lambda t} + \bar{h}_i \theta_i e^{\lambda t}}{1 + \bar{h}_i + \xi_i e^{\lambda t}} + \mathcal{O}(\xi_i \theta_i + \xi^2) \right) \\
&\quad + d \xi_j e^{\lambda t} \int_0^\infty e^{-\lambda s} g_a^p(s) ds - d \xi_i e^{\lambda t} \quad \text{Using (3)} \\
&= \frac{1}{\varepsilon} \left(\frac{\bar{p}_i \bar{h}_i}{1 + \bar{h}_i} - \frac{\bar{p}_i \bar{h}_i + \bar{p}_i \xi_i e^{\lambda t} + \bar{h}_i \theta_i e^{\lambda t}}{1 + \bar{h}_i + \xi_i e^{\lambda t}} - 2\alpha \bar{h}_i \xi_i e^{\lambda t} + \xi_i e^{\lambda t} + \mathcal{O}(\xi_i \theta_i + \xi^2) \right) + d \xi_j e^{\lambda t} \int_0^\infty e^{-\lambda s} g_a^p(s) ds - d \xi_i e^{\lambda t} \\
&= \frac{1}{\varepsilon} \left(\frac{\bar{p}_i \bar{h}_i (1 + \bar{h}_i + \xi_i e^{\lambda t}) - (\bar{p}_i \bar{h}_i + \bar{p}_i \xi_i e^{\lambda t} + \bar{h}_i \theta_i e^{\lambda t}) (1 + \bar{h}_i)}{(1 + \bar{h}_i) (1 + \bar{h}_i + \xi_i e^{\lambda t})} - 2\alpha \bar{h}_i \xi_i e^{\lambda t} + \xi_i e^{\lambda t} \right) \\
&\quad + d \xi_j e^{\lambda t} \int_0^\infty e^{-\lambda s} g_a^p(s) ds - d \xi_i e^{\lambda t} + \mathcal{O}(\xi_i \theta_i + \xi^2) \\
&= \frac{1}{\varepsilon} \left(\frac{-\bar{p}_i \xi_i e^{\lambda t} - \bar{h}_i (1 + \bar{h}_i) \theta_i e^{\lambda t}}{(1 + \bar{h}_i)^2 + (1 + \bar{h}_i) \xi_i e^{\lambda t}} + (1 - 2\alpha \bar{h}_i) \xi_i e^{\lambda t} - \varepsilon d \xi_i e^{\lambda t} + \varepsilon d \xi_j e^{\lambda t} \int_0^\infty e^{-\lambda s} g_a^p(s) ds \right) + \mathcal{O}(\xi_i \theta_i + \xi^2) \\
&= \frac{1}{\varepsilon} \left(\frac{-\bar{p}_i \xi_i e^{\lambda t} - \bar{h}_i (1 + \bar{h}_i) \theta_i e^{\lambda t} + (1 + \bar{h}_i)^2 (1 - 2\alpha \bar{h}_i) \xi_i e^{\lambda t} - \varepsilon d (1 + \bar{h}_i)^2 \xi_i e^{\lambda t}}{(1 + \bar{h}_i)^2 + (1 + \bar{h}_i) \xi_i e^{\lambda t}} \right. \\
&\quad \left. + \frac{\varepsilon d (1 + \bar{h}_i)^2 \xi_j e^{\lambda t} \int_0^\infty e^{-\lambda s} g_a^p(s) ds}{(1 + \bar{h}_i)^2 + (1 + \bar{h}_i) \xi_i e^{\lambda t}} \right) + \mathcal{O}(\xi_i^2 + \xi_i^3 + \xi_i \xi_j + \xi_i \theta_i) \\
\lambda (1 + \bar{h}_i)^2 \xi_i e^{\lambda t} &= \frac{1}{\varepsilon} \left(-\bar{p}_i \xi_i e^{\lambda t} - \bar{h}_i (1 + \bar{h}_i) \theta_i e^{\lambda t} + (1 + \bar{h}_i)^2 (1 - 2\alpha \bar{h}_i) \xi_i e^{\lambda t} - \varepsilon d (1 + \bar{h}_i)^2 \xi_i e^{\lambda t} \right. \\
&\quad \left. + \varepsilon d (1 + \bar{h}_i)^2 \xi_j e^{\lambda t} \int_0^\infty e^{-\lambda s} g_a^p(s) ds \right) + \mathcal{O}(\xi_i^2 + \xi_i^3 + \xi_i \xi_j + \xi_i \theta_i) \\
\lambda \xi_i &= \frac{1}{\varepsilon} \left(\frac{-\bar{p}_i}{(1 + \bar{h}_i)^2} \xi_i - \frac{\bar{h}_i}{1 + \bar{h}_i} \theta_i + (1 - 2\alpha \bar{h}_i) \xi_i - \varepsilon d \xi_i + \varepsilon d \xi_j \int_0^\infty e^{-\lambda s} g_a^p(s) ds \right) + \mathcal{O}(\xi_i^2 + \xi_i^3 + \xi_i \xi_j + \xi_i \theta_i) \\
\lambda \xi_i &= \frac{1}{\varepsilon} \left(\frac{-\bar{p}_i}{(1 + \bar{h}_i)^2} \xi_i - \frac{\bar{h}_i}{1 + \bar{h}_i} \theta_i + (1 - 2\alpha \bar{h}_i) \xi_i - \varepsilon d \xi_i + \varepsilon d \xi_j \frac{a^p}{(\lambda + a)^p} \right) + \mathcal{O}(\xi_i^2 + \xi_i^3 + \xi_i \xi_j + \xi_i \theta_i)
\end{aligned}$$

since

$$\int_0^\infty e^{-\lambda s} g_a^p(s) ds = \int_0^\infty \frac{a^p s^{p-1} e^{-(\lambda+a)s}}{\Gamma(p)} ds = \frac{a^p}{(\lambda + a)^p} \int_0^\infty \frac{(\lambda + a)^p s^{p-1} e^{-(\lambda+a)s}}{\Gamma(p)} ds = \frac{a^p}{(\lambda + a)^p} \int_0^\infty g_{\lambda+a}^p ds = \frac{a^p}{(\lambda + a)^p}$$

Thus, The second equation of (1) becomes

$$\begin{aligned}
\frac{d(\bar{p}_i + \theta_i e^{\lambda t})}{dt} &= \frac{(\bar{p}_i + \theta_i e^{\lambda t})(\bar{h}_i + \xi_i e^{\lambda t})}{1 + \bar{h}_i + \xi_i e^{\lambda t}} - \mu \bar{p}_i - \mu \theta_i e^{\lambda t} \\
\lambda \theta_i e^{\lambda t} &= \frac{\bar{p}_i \bar{h}_i + \bar{p}_i \xi_i e^{\lambda t} + \bar{h}_i \theta_i e^{\lambda t} + \theta_i \xi_i e^{2\lambda t}}{1 + \bar{h}_i + \xi_i e^{\lambda t}} - \frac{\bar{p}_i \bar{h}_i}{1 + \bar{h}_i} - \mu \theta_i e^{\lambda t} \quad \text{Using (3)} \\
&= \frac{(1 + \bar{h}_i)(\bar{p}_i \bar{h}_i + \bar{p}_i \xi_i e^{\lambda t} + \bar{h}_i \theta_i e^{\lambda t} + \theta_i \xi_i e^{2\lambda t}) - \bar{p}_i \bar{h}_i (1 + \bar{h}_i + \xi_i e^{\lambda t}) - \mu \theta_i e^{\lambda t} (1 + \bar{h}_i)(1 + \bar{h}_i + \xi_i e^{\lambda t})}{(1 + \bar{h}_i)(1 + \bar{h}_i + \xi_i e^{\lambda t})} \\
&= \frac{\bar{p}_i \xi_i e^{\lambda t} + (1 + \bar{h}_i)(\bar{h}_i \theta_i e^{\lambda t} + \theta_i \xi_i e^{2\lambda t}) - \mu \theta_i e^{\lambda t} (1 + \bar{h}_i)(1 + \bar{h}_i + \xi_i e^{\lambda t})}{(1 + \bar{h}_i)(1 + \bar{h}_i + \xi_i e^{\lambda t})} \\
&= \frac{\bar{p}_i \xi_i e^{\lambda t} + (1 + \bar{h}_i)[\bar{h}_i - \mu(1 + \bar{h}_i)]\theta_i e^{\lambda t}}{(1 + \bar{h}_i)^2 + (1 + \bar{h}_i)\xi_i e^{\lambda t}} + \mathcal{O}(\xi_i \theta_i) \\
\lambda(1 + \bar{h}_i)^2 \theta_i e^{\lambda t} &= \bar{p}_i \xi_i e^{\lambda t} + (1 + \bar{h}_i)[\bar{h}_i - \mu(1 + \bar{h}_i)]\theta_i e^{\lambda t} + \mathcal{O}(\xi_i \theta_i) \\
\lambda(1 + \bar{h}_i)^2 \theta_i e^{\lambda t} &= \bar{p}_i \xi_i e^{\lambda t} + (1 + \bar{h}_i)[\bar{h}_i - \mu(1 + \bar{h}_i)]\theta_i e^{\lambda t} + \mathcal{O}(\xi_i \theta_i) \\
\lambda \theta_i &= \frac{\bar{p}_i}{(1 + \bar{h}_i)^2} \xi_i + \frac{1}{(1 + \bar{h}_i)} [\bar{h}_i - \mu(1 + \bar{h}_i)] \theta_i + \mathcal{O}(\xi_i \theta_i) \\
\lambda \theta_i &= \frac{\bar{p}_i}{(1 + \bar{h}_i)^2} \xi_i + \left(\frac{\bar{h}_i}{(1 + \bar{h}_i)} - \mu \right) \theta_i + \mathcal{O}(\xi_i \theta_i)
\end{aligned}$$

Ignoring the nonlinear terms on the right hand-side of the expression of $\lambda \xi_i$ and $\lambda \theta_i$ leads to the following eigenvalue problem

$$\begin{cases} \lambda \xi_i = \frac{1}{\varepsilon} \left(1 - 2\alpha \bar{h}_i - \frac{\bar{p}_i}{(1 + \bar{h}_i)^2} - \varepsilon d \right) \xi_i + d \frac{a^p}{(\lambda + a)^p} \xi_j - \frac{1}{\varepsilon} \frac{\bar{h}_i}{1 + \bar{h}_i} \theta_i, \\ \lambda \theta_i = \frac{\bar{p}_i}{(1 + \bar{h}_i)^2} \xi_i + \left(\frac{\bar{h}_i}{1 + \bar{h}_i} - \mu \right) \theta_i, \end{cases} \quad (5)$$

for nonzero $(\xi_1, \theta_1, \xi_2, \theta_2)^T$. This eigenvalue problem can be expressed in the matrix form $A(\lambda)(\xi_1, \theta_1, \xi_2, \theta_2)^T = (0, 0, 0, 0)^T$ where

$$A(\lambda) := \left(\begin{array}{cc|cc} \frac{1}{\varepsilon} \left(1 - 2\alpha \bar{h}_1 - \frac{\bar{p}_1}{(1 + \bar{h}_1)^2} - \varepsilon d \right) - \lambda & -\frac{1}{\varepsilon} \frac{\bar{h}_1}{(1 + \bar{h}_1)} & d \frac{a^p}{(\lambda + a)^p} & 0 \\ \frac{\bar{p}_1}{(1 + \bar{h}_1)^2} & \frac{\bar{h}_1}{1 + \bar{h}_1} - \mu - \lambda & 0 & 0 \\ \hline d \frac{a^p}{(\lambda + a)^p} & 0 & \frac{1}{\varepsilon} \left(1 - 2\alpha \bar{h}_2 - \frac{\bar{p}_2}{(1 + \bar{h}_2)^2} - \varepsilon d \right) - \lambda & -\frac{1}{\varepsilon} \frac{\bar{h}_2}{(1 + \bar{h}_2)} \\ 0 & 0 & \frac{\bar{p}_2}{(1 + \bar{h}_2)^2} & \frac{\bar{h}_2}{1 + \bar{h}_2} - \mu - \lambda \end{array} \right).$$

System (5) possess a nonzero solution if and only if the determinant of the matrix $A(\lambda)$ is zero.

We assume that $\bar{h}_1 = \bar{h}_2$ and $\bar{p}_1 = \bar{p}_2$. Then the matrix $A(\lambda)$ can be rewritten as $A(\lambda) = \begin{pmatrix} A_1 - \lambda I_2 & A_2 \\ A_2 & A_1 - \lambda I_2 \end{pmatrix}$, where I_2 is the 2×2 identity matrix, A_1 and A_2 are 2×2 matrices defined by $A_1 = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ and $A_2 := \begin{pmatrix} da_{13} & 0 \\ 0 & 0 \end{pmatrix}$, with $a_{11} := \frac{1}{\varepsilon} \left(1 - 2\alpha \bar{h}_1 - \frac{\bar{p}_1}{(1 + \bar{h}_1)^2} - \varepsilon d \right)$, $a_{12} := -\frac{1}{\varepsilon} \frac{\bar{h}_1}{(1 + \bar{h}_1)}$, $a_{21} := \frac{\bar{p}_1}{(1 + \bar{h}_1)^2}$ and $a_{22} := \frac{\bar{h}_1}{1 + \bar{h}_1} - \mu$ and $a_{13} = \frac{a^p}{(\lambda + a)^p}$.

We have $\det(A) = \det(A_1 - \lambda I_2 - A_2) \times \det(A_1 - \lambda I_2 + A_2)$. Therefore, system (5) has a nonzero solution if and only if

$\det(A_1 - \lambda I_2 - A_2) = 0$ or $\det(A_1 - \lambda I_2 + A_2) = 0$. Next, we compute $\det(A_1 - \lambda I_2 \pm A_2)$. We have that:

$$\begin{aligned}\det(A_1 - \lambda I_2 \pm A_2) &= \begin{vmatrix} a_{11} - \lambda \pm a_{13} & a_{12} \\ a_{21} & a_{22} - \lambda \end{vmatrix} \\ &= (a_{11} - \lambda \pm a_{13})(a_{22} - \lambda) - a_{12}a_{21} \\ &= ((a_{11} - \lambda)(a_{22} - \lambda) - a_{12}a_{21}) \pm a_{13}(a_{22} - \lambda) \\ &= (\lambda^2 - (a_{11} + a_{22})\lambda + a_{11}a_{22} - a_{12}a_{21}) \pm a_{13}(a_{22} - \lambda)\end{aligned}$$

Therefore, $\det(A_1 - \lambda I_2 - A_2) = 0$ or $\det(A_1 - \lambda I_2 + A_2) = 0$ are equivalent to $P(\lambda) = Q(\lambda)$ or $P(\lambda) = -Q(\lambda)$, where $P(\lambda) := \lambda^2 - B\lambda + C$ and $Q(\lambda) := a_{13}(a_{22} - \lambda)$ where

$$\begin{aligned}B := a_{11} + a_{22} &= \frac{1}{\varepsilon} \left(1 - 2\alpha\bar{h}_1 - \frac{\bar{p}_1}{(1 + \bar{h}_1)^2} - \varepsilon d \right) + \frac{\bar{h}_1}{1 + \bar{h}_1} - \mu = \frac{1}{\varepsilon} \left(1 - 2\alpha\bar{h}_1 - \frac{\bar{p}_1}{(1 + \bar{h}_1)^2} \right) + \frac{\bar{h}_1}{1 + \bar{h}_1} - \mu - d \\ C := a_{11}a_{22} - a_{12}a_{21} &= \frac{1}{\varepsilon} \left(1 - 2\alpha\bar{h}_1 - \frac{\bar{p}_1}{(1 + \bar{h}_1)^2} - \varepsilon d \right) \left(\frac{\bar{h}_1}{1 + \bar{h}_1} - \mu \right) - \left(-\frac{1}{\varepsilon} \left(\frac{\bar{h}_1}{(1 + \bar{h}_1)} \right) \left(\frac{\bar{p}_1}{(1 + \bar{h}_1)^2} \right) \right) \\ &= \frac{1}{\varepsilon} \left(1 - 2\alpha\bar{h}_1 - \frac{\bar{p}_1}{(1 + \bar{h}_1)^2} - \varepsilon d \right) \left(\frac{\bar{h}_1}{1 + \bar{h}_1} - \mu \right) + \frac{1}{\varepsilon} \left(\frac{\bar{h}_1\bar{p}_1}{(1 + \bar{h}_1)^3} \right) \\ a_{13}(a_{22} - \lambda) &= d \frac{a^p}{(\lambda + a)^p} \left(\frac{\bar{h}_1}{1 + \bar{h}_1} - \mu - \lambda \right)\end{aligned}$$

Next, we study the stability of each equilibrium point of (1).

3.2 Stability of the extinction steady-state of (1)

Here $(\bar{h}_1, \bar{p}_1, \bar{h}_2, \bar{p}_2) = (0, 0, 0, 0)$. We have $B = \frac{1}{\varepsilon} - \mu - d$ and $C = -\mu \left(\frac{1}{\varepsilon} - d \right)$. Thus, $P(\lambda) = \lambda^2 - \left(\frac{1}{\varepsilon} - \mu - d \right) \lambda - \mu \left(\frac{1}{\varepsilon} - d \right) = (\lambda + \mu) \left(\lambda + d - \frac{1}{\varepsilon} \right)$ and $Q(\lambda) = (-\mu - \lambda) d \frac{a^p}{(\lambda + a)^p}$. Therefore,

$$\begin{aligned}P(\lambda) \pm Q(\lambda) = 0 &\implies (\lambda + \mu) \left(\lambda + d - \frac{1}{\varepsilon} \right) \pm (\mu + \lambda) d \left(\frac{a^p}{(\lambda + a)^p} \right) = 0 \\ &\implies (\lambda + \mu) \left[\left(\lambda + d - \frac{1}{\varepsilon} \right) \pm d \left(\frac{a}{\lambda + a} \right)^p \right] = 0 \\ &\implies \lambda = -\mu \quad \text{or} \quad f_{\pm}(\lambda) := \left(\lambda + d - \frac{1}{\varepsilon} \right) \pm d \left(\frac{a}{\lambda + a} \right)^p = 0\end{aligned}$$

Since μ is positive, the condition for the stability is that all the roots of equation $f \pm (\lambda) = 0$ have negative real part. We have that: $f_-(0) = d - \frac{1}{\varepsilon} - d = -\frac{1}{\varepsilon} < 0$ and as $\lambda \rightarrow \infty$, $f_-(\lambda) \rightarrow \infty$. Therefore, by the intermediate value theorem there exists $\lambda > 0$ such that $f_-(\lambda) = 0$. Thus, $(0, 0, 0, 0)$ is unstable.

We observe that $f_+(0) = 2d - \frac{1}{\varepsilon}$ take the value zero when $d = \frac{1}{2\varepsilon}$. For $d = \frac{1}{2\varepsilon}$, the solution of the steady state model (3) satisfies $\bar{h}_1 + \bar{h}_2 = 0$ with $\bar{p}_1 = \bar{p}_2 = 0$. So, solution which bifurcates from $(0, 0, 0, 0)$ has negative component. We are not interested on them.

3.3 Stability of the equilibrium point $(\bar{h}_1, \bar{p}_1, \bar{h}_2, \bar{p}_2) = \left(\frac{1}{\alpha}, 0, \frac{1}{\alpha}, 0 \right)$ of (1)

At $(\bar{h}_1, \bar{p}_1, \bar{h}_2, \bar{p}_2) = \left(\frac{1}{\alpha}, 0, \frac{1}{\alpha}, 0 \right)$, we have $\frac{\bar{h}_1}{1 + \bar{h}_1} = \frac{1}{1 + \alpha}$ and $Q(\lambda) = \left(\frac{1}{1 + \alpha} - \mu - \lambda \right) d \left(\frac{a}{\lambda + a} \right)^p$ and

$$P(\lambda) = \lambda^2 - \left(\frac{1}{1 + \alpha} - \frac{1}{\varepsilon} - \mu - d \right) \lambda - \left(\frac{1}{1 + \alpha} - \mu \right) \left(\frac{1}{\varepsilon} + d \right) = \left(\lambda + d + \frac{1}{\varepsilon} \right) \left(\lambda + \mu - \frac{1}{1 + \alpha} \right).$$

Therefore,

$$\begin{aligned}
P(\lambda) \pm Q(\lambda) &\implies \left(\lambda + d + \frac{1}{\varepsilon}\right) \left(\lambda + \mu - \frac{1}{1+\alpha}\right) \pm \left(\frac{1}{1+\alpha} - \mu - \lambda\right) d \left(\frac{a}{\lambda+a}\right)^p = 0 \\
&\implies \left(\lambda + \mu - \frac{1}{1+\alpha}\right) \left[\left(\lambda + d + \frac{1}{\varepsilon}\right) \pm d \left(\frac{a}{\lambda+a}\right)^p\right] = 0 \\
&\implies \lambda = \frac{1}{1+\alpha} - \mu \quad \text{or} \quad f_{\pm}(\lambda) := \left(\lambda + d + \frac{1}{\varepsilon}\right) \pm d \left(\frac{a}{\lambda+a}\right)^p = 0
\end{aligned}$$

We have that $f_{\pm}(0) = d + \frac{1}{\varepsilon} \pm d > 0$. Hence $\lambda = 0$ is not a solution of $f_{\pm}(\lambda) = 0$. Let's define $\gamma = d + \frac{1}{\varepsilon}$ and use the expression $a = \frac{p}{\tau}$, where τ is the mean of the gamma distribution, $f_{\pm}$ becomes $f_{\pm}(\lambda) = \lambda + d + \frac{1}{\varepsilon} \pm d \left(\frac{1}{1 + \frac{\tau}{p}\lambda}\right)^p$. Thus, the equation $f_{\pm}(\lambda) = 0$ becomes

$$\left(\lambda + d + \frac{1}{\varepsilon}\right) \left(1 + \frac{\tau}{p}\lambda\right)^p \pm d = 0, \quad p \in \mathbb{R} \quad (6)$$

We want to investigate the sign of the real part of the roots of equation (6) as the mean delay τ and parameter p varies.

- For $p = 1$, equation (6) gives $\left(\lambda + d + \frac{1}{\varepsilon}\right) (1 + \tau\lambda) \pm d = 0$, that is $\tau\lambda^2 + \left(1 + \tau\left(d + \frac{1}{\varepsilon}\right)\right)\lambda + \left(d + \frac{1}{\varepsilon} \pm d\right) = 0$. We note

that $d + \frac{1}{\varepsilon} \pm d > 0$. The roots of this equation are $\lambda_{\pm} = \frac{-\left(1 + \tau\left(d + \frac{1}{\varepsilon}\right)\right) \pm \sqrt{\Delta}}{2\tau}$, where $\Delta = \left(1 + \tau\left(d + \frac{1}{\varepsilon}\right)\right)^2 - 4(\gamma \pm d)$.

If $\Delta \leq 0$ then $\text{Re}(\lambda_{\pm}) = \frac{-\left(1 + \tau\left(d + \frac{1}{\varepsilon}\right)\right)}{2\tau} < 0$. If $\Delta > 0$, since $\gamma \pm d > 0$, we have $\sqrt{\Delta} < \left|1 + \tau\left(d + \frac{1}{\varepsilon}\right)\right|$, thus $\lambda_{\pm} < 0$.

- For $p \rightarrow \infty$, equation (6) goes to $\left(\lambda + d + \frac{1}{\varepsilon}\right) e^{\tau\lambda} \pm d = 0$, which is equivalent to the equation

$$\lambda + d + \frac{1}{\varepsilon} \pm d e^{-\tau\lambda} = 0. \quad (7)$$

The stability conditions of this transcendental equation (7) are $-\left(d + \frac{1}{\varepsilon}\right) + d < 0$ and $-\left(d + \frac{1}{\varepsilon}\right) - d < 0$. Since $-\left(d + \frac{1}{\varepsilon}\right) + d = -\frac{1}{\varepsilon} < 0$ and $-\left(d + \frac{1}{\varepsilon}\right) - d = -\left(2d + \frac{1}{\varepsilon}\right) < 0$, we conclude that all solutions of (7) have negative real parts.

- For $\tau = 0$, equation (6) gives $\lambda + d + \frac{1}{\varepsilon} \pm d = 0$, that is $\lambda = -\left(d + \frac{1}{\varepsilon} \pm d\right) < 0$.
- Now we assume $p > 0$ and $\tau > 0$. We want to show that equation (6) does not have a purely imaginary solution. Let $\lambda = i\omega$, $\omega > 0$. Equation (6) becomes

$$\left(d + \frac{1}{\varepsilon} + i\omega\right) \left(1 + i\frac{\tau}{p}\omega\right)^p \pm d = 0. \quad (8)$$

In the polar form

$$\begin{aligned}
(8) &\implies \left(\left(d + \frac{1}{\varepsilon}\right)^2 + \omega^2\right)^{1/2} e^{i\theta_1} \left(1 + \left(\frac{\tau}{p}\omega\right)^2\right)^{p/2} e^{ip\theta_2} = \pm d \\
&\implies \left[\left(\left(d + \frac{1}{\varepsilon}\right)^2 + \omega^2\right) \left(1 + \left(\frac{\tau}{p}\omega\right)^2\right)^p\right]^{1/2} e^{i(\theta_1 + p\theta_2)} = \pm d
\end{aligned}$$

$$\Rightarrow \begin{cases} \left[\left(\left(d + \frac{1}{\varepsilon} \right)^2 + \omega^2 \right) \left(1 + \left(\frac{\tau}{p} w \right)^2 \right)^p \right]^{1/2} \cos(\theta_1 + p\theta_2) = \pm d & (9a) \\ \left[\left(\left(d + \frac{1}{\varepsilon} \right)^2 + \omega^2 \right) \left(1 + \left(\frac{\tau}{p} w \right)^2 \right)^p \right]^{1/2} \sin(\theta_1 + p\theta_2) = 0 & (9b) \end{cases}$$

Adding the squares of (9a) and (9b), we obtain

$$\left(\left(d + \frac{1}{\varepsilon} \right)^2 + \omega^2 \right) \left(1 + \left(\frac{\tau}{p} w \right)^2 \right)^p = d^2 \quad (10)$$

If p is a positive integer, then $\left(1 + \left(\frac{\tau}{p} w \right)^2 \right)^p = \sum_{k=0}^p \binom{p}{k} \left(\frac{\tau^2}{p^2} w^2 \right)^k$, where $k < p$ is a positive integer. Hence,

$$\begin{aligned} (10) &\Rightarrow \left(\left(d + \frac{1}{\varepsilon} \right)^2 + \omega^2 \right) \sum_{k=0}^p \binom{p}{k} \left(\frac{\tau^2}{p^2} w^2 \right)^k = d^2 \\ &\Rightarrow \sum_{k=0}^p \binom{p}{k} \left(\frac{\tau^2}{p^2} \right)^k \left(\left(d + \frac{1}{\varepsilon} \right)^2 + \omega^2 \right) (w^2)^k = d^2 \\ &\Rightarrow \sum_{k=0}^p \binom{p}{k} \left(\frac{\tau^2}{p^2} \right)^k \left(\left(d + \frac{1}{\varepsilon} \right)^2 (w^2)^k + (w^2)^{k+1} \right) = d^2 \\ &\Rightarrow \sum_{k=0}^p \binom{p}{k} \left(\frac{\tau^2}{p^2} \right)^k \left(\left(d + \frac{1}{\varepsilon} \right)^2 W^k + W^{k+1} \right) = d^2, \quad \text{where } 0 < W := w^2 \\ &\Rightarrow \sum_{k=0}^p b_k \left(\left(d + \frac{1}{\varepsilon} \right)^2 W^k + W^{k+1} \right) - d^2 = 0, \quad \text{where } b_k = \binom{p}{k} \left(\frac{\tau^2}{p^2} \right)^k \\ &\Rightarrow b_0 \left(\left(d + \frac{1}{\varepsilon} \right)^2 + W \right) - d^2 + b_1 \left(\left(d + \frac{1}{\varepsilon} \right)^2 W + W^2 \right) + \dots + b_{p-1} \left(\left(d + \frac{1}{\varepsilon} \right)^2 W^{p-1} + W^p \right) \\ &\quad + b_p \left(\left(d + \frac{1}{\varepsilon} \right)^2 W^p + W^{p+1} \right) = 0 \\ &\Rightarrow \left(b_0 \left(d + \frac{1}{\varepsilon} \right)^2 - d^2 \right) + b_0 W + \left(b_1 \left(d + \frac{1}{\varepsilon} \right)^2 W + b_1 W^2 \right) + \dots + \left(b_{p-1} \left(d + \frac{1}{\varepsilon} \right)^2 W^{p-1} + b_{p-1} W^p \right) \\ &\quad + \left(b_p \left(d + \frac{1}{\varepsilon} \right)^2 W^p + b_p W^{p+1} \right) = 0 \\ &\Rightarrow \left(b_0 \left(d + \frac{1}{\varepsilon} \right)^2 - d^2 \right) + \left(b_0 + b_1 \left(d + \frac{1}{\varepsilon} \right)^2 \right) W + \left(b_1 + b_2 \left(d + \frac{1}{\varepsilon} \right)^2 \right) W^2 \\ &\quad + \dots + \left(b_{p-1} + b_p \left(d + \frac{1}{\varepsilon} \right)^2 \right) W^p + b_p W^{p+1} = 0 \end{aligned}$$

The latter equation can be rewritten as

$$\beta_{p+1} W^{p+1} + \beta_p W^p + \dots + \beta_2 W^2 + \beta_1 W + \beta_0 = 0 \quad (11)$$

where $\beta_{p+1} := b_p = \left(\frac{\tau^2}{p^2} \right)^p > 0$, $\beta_0 := b_0 \left(d + \frac{1}{\varepsilon} \right)^2 - d^2 = \left(d + \frac{1}{\varepsilon} \right)^2 - d^2 = \frac{1}{\varepsilon} \left(2d + \frac{1}{\varepsilon} \right) > 0$ and $\beta_k := b_{k-1} + b_k \left(d + \frac{1}{\varepsilon} \right)^2 > 0$, $k = 1, \dots, p$.

Since the coefficients β_k , $k = 0, \dots, p+1$ are all positive, equation (11) has no positive solution $W = \omega^2$. That is, the equation (6) does not have a purely imaginary root $\lambda = i\omega$ with $\omega > 0$. Therefore, all the roots have a negative real part when $\tau > 0$ and p is a positive integer.

If p is any real number, then $\left(1 + \left(\frac{\tau}{p}w\right)^2\right)^p = \sum_{k \geq 0} \binom{p}{k} \left(\frac{\tau^2}{p^2}w^2\right)^k$, where the sum is taken over all positive integers k .

Hence

$$\begin{aligned}
(10) &\Rightarrow \left(\left(d + \frac{1}{\varepsilon}\right)^2 + \omega^2\right) \sum_{k \geq 0} \binom{p}{k} \left(\frac{\tau^2}{p^2}w^2\right)^k = d^2 \\
&\Rightarrow \sum_{k \geq 0} b_k \left(\left(d + \frac{1}{\varepsilon}\right)^2 W^k + W^{k+1}\right) - d^2 = 0 \\
&\Rightarrow b_0 \left(\left(d + \frac{1}{\varepsilon}\right)^2 + W\right) - d^2 + b_1 \left(\left(d + \frac{1}{\varepsilon}\right)^2 W + W^2\right) + b_2 \left(\left(d + \frac{1}{\varepsilon}\right)^2 W^2 + W^3\right) + \dots = 0 \\
&\Rightarrow \left(b_0 \left(d + \frac{1}{\varepsilon}\right)^2 - d^2\right) W^0 + \left(b_0 + b_1 \left(d + \frac{1}{\varepsilon}\right)^2\right) W + \left(b_1 + b_2 \left(d + \frac{1}{\varepsilon}\right)^2\right) W^2 + \dots = 0
\end{aligned}$$

Since the terms of the infinite series are all positive, it cannot be equal to zero. So, we cannot find a positive W such that the latter equation is satisfied. That is, equation (6) does not have a purely imaginary root $\lambda = i\omega$ with $\omega > 0$. Therefore, all the roots have a negative real part when $\tau > 0$ and p is any real number.

From the condition $\lambda = \frac{1}{1+\alpha} - \mu$, we have $\lambda < 0$ if $\mu > \frac{1}{1+\alpha}$, $\lambda > 0$ if $\mu < \frac{1}{1+\alpha}$ and $\lambda = 0$ if $\mu = \frac{1}{1+\alpha}$.

We then conclude that the equilibrium point $(\bar{h}_1, \bar{p}_1, \bar{h}_2, \bar{p}_2) = \left(\frac{1}{\alpha}, 0, \frac{1}{\alpha}, 0\right)$ is stable if $\mu > \frac{1}{1+\alpha}$ and unstable if $\mu < \frac{1}{1+\alpha}$.

At the bifurcation point $\mu = \frac{1}{1+\alpha}$, we have one zero eigenvalue and all the other have negative real parts. So, for a small perturbation from the bifurcation point, the eigenvalues with negative real parts remains negative and the zero eigenvalue will determine the stability close to the bifurcation point.

3.4 Stability of the coexistence equilibrium of (1)

Here $(\bar{h}_1, \bar{h}_2, \bar{p}_1, \bar{p}_2) = (\bar{h}_1, \bar{h}_1, \bar{p}_1, \bar{p}_1)$ with $\bar{h}_1 = \frac{\mu}{1-\mu}$ and $\bar{p}_1 = \frac{1-\mu(1+\alpha)}{(1-\mu)^2}$. We have that

$$1 + \bar{h}_1 = \frac{1}{1-\mu}; \frac{\bar{h}_1}{1 + \bar{h}_1} = \mu; \bar{p}_1 \bar{h}_1 = \frac{\mu - \mu^2(1+\alpha)}{(1-\mu)^3}; \frac{\bar{p}_1 \bar{h}_1}{(1 + \bar{h}_1)^3} = \mu - \mu^2(1+\alpha); \frac{\bar{p}_1}{(1 + \bar{h}_1)^2} = 1 - \mu(1+\alpha).$$

Then,

$$\begin{aligned}
B &= \frac{1}{\varepsilon} \left(1 - 2\alpha\bar{h}_1 - \frac{\bar{p}_1}{(1 + \bar{h}_1)^2}\right) + \frac{\bar{h}_1}{1 + \bar{h}_1} - \mu - d = \frac{1}{\varepsilon} \left(1 - 2\alpha\frac{\mu}{1-\mu} - 1 + \mu(1+\alpha)\right) + \mu - \mu - d \\
&= \frac{1}{\varepsilon} \left(-2\alpha\frac{\mu}{1-\mu} + \mu(1+\alpha)\right) - d = \frac{\mu}{\varepsilon} \left((1+\alpha) - \frac{2\alpha}{1-\mu}\right) - d = \frac{\mu}{\varepsilon(1-\mu)} [(1+\alpha)(1-\mu) - 2\alpha] - d \\
&= \frac{\mu}{\varepsilon(1-\mu)} (1 - \mu + \alpha - \mu\alpha - 2\alpha) - d = \frac{\mu}{\varepsilon(1-\mu)} (1 - \mu - \alpha - \mu\alpha) - d = \frac{\mu}{\varepsilon(1-\mu)} (1 - \alpha - \mu(1+\alpha)) - d \\
C &= \frac{1}{\varepsilon} \left(1 - 2\alpha\bar{h}_1 - \frac{\bar{p}_1}{(1 + \bar{h}_1)^2} - \varepsilon d\right) \left(\frac{\bar{h}_1}{1 + \bar{h}_1} - \mu\right) + \frac{1}{\varepsilon} \left(\frac{\bar{h}_1 \bar{p}_1}{(1 + \bar{h}_1)^3}\right) \\
&= \frac{1}{\varepsilon} \left(1 - 2\alpha\bar{h}_1 - \frac{\bar{p}_1}{(1 + \bar{h}_1)^2} - \varepsilon d\right) (\mu - \mu) + \frac{1}{\varepsilon} \left(\frac{\bar{h}_1 \bar{p}_1}{(1 + \bar{h}_1)^3}\right) = \frac{1}{\varepsilon} \left(\frac{\bar{h}_1 \bar{p}_1}{(1 + \bar{h}_1)^3}\right) = \frac{1}{\varepsilon} (\mu - \mu^2(1+\alpha)) \\
&= \frac{1}{\varepsilon} \mu (1 - \mu(1+\alpha))
\end{aligned}$$

We observe that B can be written as $\frac{C - \alpha \frac{\mu}{\varepsilon}}{1 - \mu} - d$. Therefore, $P(\lambda) = \lambda^2 + \left(d - \frac{C - \alpha \frac{\mu}{\varepsilon}}{1 - \mu}\right) \lambda + C$ and $Q(\lambda) = -d\lambda \left(\frac{a}{\lambda + a}\right)^p = -d\lambda \left(\frac{1}{1 + \frac{\tau}{p}\lambda}\right)^p$. The characteristic equation $P(\lambda) \pm Q(\lambda) = 0$, reads

$$f_{\pm}(\lambda) := \left(\lambda^2 + \left(d - \frac{C - \alpha \frac{\mu}{\varepsilon}}{1 - \mu}\right) \lambda + C\right) \left(1 + \frac{\tau}{p}\lambda\right)^p \pm d\lambda = 0. \quad (12)$$

$\lambda = 0$ is a solution of (12) if $C = 0$, that is if $\mu = \frac{1}{1 + \alpha}$, which is the bifurcation point for the equilibrium point $(\bar{h}_1, \bar{p}_1, \bar{h}_2, \bar{p}_2) = \left(\frac{1}{\alpha}, 0, \frac{1}{\alpha}, 0\right)$.

Let's analyze the stability of this equilibrium point $(\bar{h}_1, \bar{h}_2, \bar{p}_1, \bar{p}_2) = (\bar{h}_1, \bar{h}_1, \bar{p}_1, \bar{p}_1)$ close to the bifurcation point, that is, when $\mu = \frac{1}{1 + \alpha} - \delta$, where $\delta > 0$ is small enough. In this case, $C = \frac{1}{\varepsilon} \left(\frac{1}{1 + \alpha} - \delta\right) \left(1 - \left(\frac{1}{1 + \alpha} - \delta\right)(1 + \alpha)\right) = \frac{\delta(1 + \alpha)}{\varepsilon} \left(\frac{1}{1 + \alpha} - \delta\right) = \frac{\delta - \delta^2(1 + \alpha)}{\varepsilon}$ and at $\delta = 0$, $\lambda(0) = 0$.

We have

$$f_{\pm}(\delta, \lambda(\delta)) = \left(\lambda^2(\delta) + \left(d - (1 + \alpha) \frac{\delta - \delta^2(1 + \alpha) - \frac{\alpha}{1 + \alpha} - \alpha\delta}{\alpha\varepsilon}\right) \lambda(\delta) + \frac{\delta - \delta^2(1 + \alpha)}{\varepsilon}\right) \left(1 + \frac{\tau}{p}\lambda\right)^p \pm d\lambda(\delta).$$

We have $f_{\pm}(0, 0) = 0$ and $\left.\frac{\partial f}{\partial \lambda}\right|_{(0,0)} = \left(d - (1 + \alpha) \frac{\delta - \delta^2(1 + \alpha) - \frac{\alpha}{1 + \alpha} - \alpha\delta}{\alpha\varepsilon}\right) \pm d \Big|_{(0,0)} = \frac{1}{\varepsilon} + d \pm d > 0$. By the implicit

function theorem, there exists a neighborhood of $(0, 0)$ so that whenever (δ, λ) is sufficiently close to $(0, 0)$, there exists a unique $\mathcal{C}^1$ -function $\lambda(\lambda)$ satisfying $f_{\pm}(\delta, \lambda(\delta)) = 0$.

Differentiating both sides of $f_{\pm}(\delta, \lambda(\delta)) = 0$ with respect to λ gives $\frac{\partial f_{\pm}}{\partial \delta} + \frac{d\lambda}{d\delta} \cdot \frac{\partial f_{\pm}}{\partial \lambda} = 0$ and solving for $\frac{d\lambda}{d\delta}$ leads us to $\frac{d\lambda}{d\delta} = -\frac{\partial f_{\pm}}{\partial \delta} / \frac{\partial f_{\pm}}{\partial \lambda}$. We have $\left.\frac{\partial f}{\partial \delta}\right|_{(0,0)} = \frac{1 - 2\delta(1 + \alpha)}{\varepsilon} \Big|_{(0,0)} = \frac{1}{\varepsilon}$. Therefore $\frac{d\lambda}{d\delta}(0) = -\frac{1}{\varepsilon} / \left(\frac{1}{\varepsilon} + d \pm d\right) < 0$.

The Taylor series of $\lambda(\delta)$ at $\delta = 0$ is then

$$\lambda(\delta) = \lambda(0) + \delta \frac{d\lambda}{d\delta}(0) + \mathcal{O}(\delta^2) = -\frac{1}{\varepsilon} / \left(\frac{1}{\varepsilon} + d \pm d\right) + \mathcal{O}(\delta^2).$$

Therefore, $\lambda = 0$ is not an eigenvalue when $\delta > 0$. Therefore $\lambda(\delta) < 0$ for $0 < \delta \ll 1$. Since at the bifurcation point $\mu = \frac{1}{1 + \alpha}$, all the other eigenvalues of the equilibrium point $(\bar{h}_1, \bar{p}_1, \bar{h}_2, \bar{p}_2) = \left(\frac{1}{\alpha}, 0, \frac{1}{\alpha}, 0\right)$ have negative real parts, they would remain negative for δ sufficiently small.

We conclude that when $\mu < \frac{1}{1 + \alpha}$, the equilibrium point $(\bar{h}_1, \bar{h}_1, \bar{p}_1, \bar{p}_1)$ with $\bar{h}_1 = \frac{\mu}{1 - \mu}$ and $\bar{p}_1 = \frac{1 - \mu(1 + \alpha)}{(1 - \mu)^2}$ is stable.

3.4.1 Hopf bifurcation when $\tau = 0$

Here we look at the Hopf bifurcation for $\tau = 0$ and $\tau > 0$. Assume that $\mu < \frac{1}{1 + \alpha}$ and suppose that $\lambda = i\omega$, $\omega > 0$ is a root of (12). We have for fixed p that:

$$\left[\left(C - \omega^2 + i \left(d - \frac{C - \alpha \frac{\mu}{\varepsilon}}{1 - \mu} \right) \omega \right) \left(1 + i \frac{\tau}{p} \omega \right)^p - id\omega \right] \left[\left(C - \omega^2 + i \left(d - \frac{C - \alpha \frac{\mu}{\varepsilon}}{1 - \mu} \right) \omega \right) \left(1 + i \frac{\tau}{p} \omega \right)^p + id\omega \right] = 0. \quad (13)$$

When $\tau = 0$, equation (13) becomes $C - \omega^2 + i\omega \left(d - \frac{C - \alpha \frac{\mu}{\varepsilon}}{1 - \mu} \pm d \right) = 0$, which implies

$$\begin{cases} C - \omega^2 = 0 \end{cases} \quad (14a)$$

$$\begin{cases} d - \frac{C - \alpha \frac{\mu}{\varepsilon}}{1 - \mu} \pm d = 0. \end{cases} \quad (14b)$$

Step 1. Next, we solve (14) for μ and ω .

- From (14a), we obtain $\omega^2 = C = \frac{1}{\varepsilon} \mu (1 - \mu(1 + \alpha))$, which is positive since $\mu < \frac{1}{1 + \alpha}$.
- Equation (14b) with the minus sign gives $\frac{C - \alpha \frac{\mu}{\varepsilon}}{1 - \mu} = 0$, that is $\frac{1}{\varepsilon} \mu (1 - \alpha - \mu(1 + \alpha)) = 0$, that is $\mu = \frac{1 - \alpha}{1 + \alpha}$. If $\mu = \frac{1 - \alpha}{1 + \alpha}$ then $\omega^2 = \frac{1}{\varepsilon} \left(\frac{1 - \alpha}{1 + \alpha} \right) \left(1 - \left(\frac{1 - \alpha}{1 + \alpha} \right) (1 + \alpha) \right) = \frac{\alpha(1 - \alpha)}{\varepsilon(1 + \alpha)}$.
- Equation (14b) with the plus sign is $d - \frac{C - \alpha \frac{\mu}{\varepsilon}}{1 - \mu} + d = 0$, that is $2d - \frac{\frac{1}{\varepsilon} \mu (1 - \alpha - \mu(1 + \alpha))}{1 - \mu} = 0$. We have that

$$\begin{aligned} 2d - \frac{\frac{1}{\varepsilon} \mu (1 - \alpha - \mu(1 + \alpha))}{1 - \mu} = 0 &\implies 2d(1 - \mu) - \frac{1}{\varepsilon} \mu (1 - \alpha - \mu(1 + \alpha)) = 0 \implies 2\varepsilon d(1 - \mu) - \mu(1 - \alpha) + \mu^2(1 + \alpha) = 0 \\ &\implies (1 + \alpha)\mu^2 - \mu(2\varepsilon d + 1 - \alpha) + 2\varepsilon d = 0 \end{aligned} \quad (15)$$

The discriminant of this latter equation is

$$\begin{aligned} \Delta &= (2\varepsilon d + 1 - \alpha)^2 - 8\varepsilon d(1 + \alpha) = (2\varepsilon d)^2 + (1 - \alpha)^2 + 4\varepsilon d(1 - \alpha) - 8\varepsilon d(1 + \alpha) = (2\varepsilon d)^2 + (1 - \alpha)^2 + 4\varepsilon d(1 - \alpha - 2 - 2\alpha) \\ &= (2\varepsilon d)^2 + (1 - \alpha)^2 - 4\varepsilon d(1 + 3\alpha) = (2\varepsilon d)^2 - 2(1 + 3\alpha)(2\varepsilon d) + (1 - \alpha)^2 \end{aligned}$$

Let's factorize this discriminant by finding the roots of the equation $x^2 - 2(1 + 3\alpha)x + (1 - \alpha)^2 = 0$ where $x = 2\varepsilon d$. The discriminant of this latter equation is $\Delta_1 = 4(1 + 3\alpha)^2 - 4(1 - \alpha)^2 = 4(1 + 3\alpha - 1 + \alpha)(1 + 3\alpha + 1 - \alpha) = 4(4\alpha)(2 + 2\alpha) = 16\alpha(2 + 2\alpha) > 0$. The roots are

$$x_{\pm} = \frac{2(1 + 3\alpha) \pm 2\sqrt{(1 + 3\alpha)^2 - (1 - \alpha)^2}}{2} = (1 + 3\alpha) \pm \sqrt{(1 + 3\alpha)^2 - (1 - \alpha)^2} = (1 + 3\alpha) \pm \sqrt{8\alpha + 8\alpha^2}.$$

Since $\Delta_1 > 0$, we have $\sqrt{\Delta} \leq |1 + 3\alpha| = 1 + 3\alpha$. Hence $x_{\pm}$ is positive with $x_- < x_+$. Therefore, $\Delta = (2\varepsilon d - x_-)(2\varepsilon d - x_+)$ and is positive if $2\varepsilon d < x_-$ or $2\varepsilon d > x_+$ and zero if $2\varepsilon d = x_-$ and the solutions of (15) reads

$$\mu_{\pm} = \frac{2\varepsilon d + 1 - \alpha \pm \sqrt{(2\varepsilon d - (1 + 3\alpha) + \sqrt{8\alpha + 8\alpha^2})(2\varepsilon d - (1 + 3\alpha) - \sqrt{8\alpha + 8\alpha^2})}}{2(1 + \alpha)} \quad (16)$$

- If $2\varepsilon d = 1 + 3\alpha - \sqrt{8\alpha + 8\alpha^2}$ then $\Delta = 0$ and $\mu_{\pm} = \frac{2\varepsilon d + 1 - \alpha}{2(1 + \alpha)} = \frac{1 + 3\alpha - \sqrt{8\alpha + 8\alpha^2} + 1 - \alpha}{2(1 + \alpha)} = \frac{2 + 2\alpha - 2\sqrt{2\alpha + 2\alpha^2}}{2(1 + \alpha)} = \frac{1 + \alpha - \sqrt{2\alpha(1 + \alpha)}}{1 + \alpha} < \frac{1}{1 + \alpha}$.
- If $2\varepsilon d = 1 + 3\alpha + \sqrt{8\alpha + 8\alpha^2}$ then $\Delta = 0$ and $\mu_{\pm} = \frac{2\varepsilon d + 1 - \alpha}{2(1 + \alpha)} = \frac{1 + 3\alpha + \sqrt{8\alpha + 8\alpha^2} + 1 - \alpha}{2(1 + \alpha)} = \frac{2 + 2\alpha + 2\sqrt{2\alpha + 2\alpha^2}}{2(1 + \alpha)} = \frac{1 + \alpha + \sqrt{2\alpha(1 + \alpha)}}{1 + \alpha} > \frac{1}{1 + \alpha}$.
- If $2\varepsilon d < 1 + 3\alpha - \sqrt{8\alpha + 8\alpha^2}$ then $\Delta > 0$ and $\mu_{\pm} < \frac{1}{1 + \alpha}$.
- If $2\varepsilon d > 1 + 3\alpha + \sqrt{8\alpha + 8\alpha^2}$ then $\Delta > 0$ and $\mu_{\pm} > \frac{1}{1 + \alpha}$.

Hence from equation (14b), we obtain $\mu = \mu_{\pm}$, defined by (16), provided that $2\varepsilon d \leq 1 + 3\alpha - \sqrt{8\alpha + 8\alpha^2}$.

In summary, when $\tau = 0$, $\lambda = i\omega$ is a solution of the characteristic equation (12) when $\mu = \frac{1-\alpha}{1+\alpha}$ and $\mu_{\pm} = \frac{2\varepsilon d + 1 - \alpha \pm \sqrt{(2\varepsilon d - (1 + 3\alpha) + \sqrt{8\alpha + 8\alpha^2})}}{2(1 + \alpha)}$ provided that $2\varepsilon d \leq 1 + 3\alpha - \sqrt{8\alpha + 8\alpha^2}$.

Step 2. Next, we compare these three values of μ to order them. We have: $\mu_- < \mu_+$. Let's show that $\mu_+ < \mu$. We have that:

$$\begin{aligned}
0 < 16\alpha\varepsilon d &\implies -12\alpha\varepsilon d < 4\alpha\varepsilon d \\
&\implies -12\alpha\varepsilon d + 1 < 4\alpha\varepsilon d + 1 \\
&\implies -12\alpha\varepsilon d + 1 + 6\alpha + 9\alpha^2 - 8\alpha - 8\alpha^2 < 4\alpha\varepsilon d + 1 - 2\alpha + \alpha^2 \\
&\implies -12\alpha\varepsilon d + (1 + 3\alpha)^2 - (8\alpha + 8\alpha^2) < 4\alpha\varepsilon d + (1 - \alpha)^2 \\
&\implies -4\varepsilon d - 12\alpha\varepsilon d + (1 + 3\alpha)^2 - (8\alpha + 8\alpha^2) < -4\varepsilon d + 4\alpha\varepsilon d + (1 - \alpha)^2 \\
&\implies 2\varepsilon d(-2(1 + 3\alpha)) + (1 + 3\alpha)^2 - (8\alpha + 8\alpha^2) < 2\varepsilon d(-2(1 - \alpha)) + (1 - \alpha)^2 \\
&\implies 4\varepsilon^2 d^2 - 2(2\varepsilon d)(1 + 3\alpha) + (1 + 3\alpha)^2 - (8\alpha + 8\alpha^2) < 4\varepsilon^2 d^2 + 2\varepsilon d(-2(1 - \alpha)) + (1 - \alpha)^2 \\
&\implies \left(2\varepsilon d - (1 + 3\alpha) + \sqrt{8\alpha + 8\alpha^2}\right) \left(2\varepsilon d - (1 + 3\alpha) - \sqrt{8\alpha + 8\alpha^2}\right) < (2\varepsilon d - (1 - \alpha))^2 \\
&\implies \sqrt{\left(2\varepsilon d - (1 + 3\alpha) + \sqrt{8\alpha + 8\alpha^2}\right) \left(2\varepsilon d - (1 + 3\alpha) - \sqrt{8\alpha + 8\alpha^2}\right)} < 1 - \alpha - 2\varepsilon d. \tag{17}
\end{aligned}$$

We will return to (17) soon. We show that $1 - \alpha - 2\varepsilon d > 0$. To do so, we notice that $\alpha < 1$ implies $4\alpha = \sqrt{8\alpha^2 + 8\alpha^2} < \sqrt{8\alpha + 8\alpha^2}$. Since $2\varepsilon d \leq 1 + 3\alpha - \sqrt{8\alpha + 8\alpha^2}$, $1 - \alpha - 2\varepsilon d \geq \sqrt{8\alpha + 8\alpha^2} - 4\alpha > 0$. Since the right hand side of (17) is positive while its left hand side also positive by condition on $2\varepsilon d$, we have

$$\begin{aligned}
&\sqrt{\left(2\varepsilon d - (1 + 3\alpha) + \sqrt{8\alpha + 8\alpha^2}\right) \left(2\varepsilon d - (1 + 3\alpha) - \sqrt{8\alpha + 8\alpha^2}\right)} < 1 - \alpha - 2\varepsilon d \\
2\varepsilon d + \sqrt{\left(2\varepsilon d - (1 + 3\alpha) + \sqrt{8\alpha + 8\alpha^2}\right) \left(2\varepsilon d - (1 + 3\alpha) - \sqrt{8\alpha + 8\alpha^2}\right)} &< 1 - \alpha \\
2\varepsilon d + 1 - \alpha + \sqrt{\left(2\varepsilon d - (1 + 3\alpha) + \sqrt{8\alpha + 8\alpha^2}\right) \left(2\varepsilon d - (1 + 3\alpha) - \sqrt{8\alpha + 8\alpha^2}\right)} &< 2 - 2\alpha \\
\frac{2\varepsilon d + 1 - \alpha + \sqrt{\left(2\varepsilon d - (1 + 3\alpha) + \sqrt{8\alpha + 8\alpha^2}\right) \left(2\varepsilon d - (1 + 3\alpha) - \sqrt{8\alpha + 8\alpha^2}\right)}}{2(1 + \alpha)} &< \frac{1 - \alpha}{1 + \alpha}
\end{aligned}$$

In conclusion, we have $\mu_- < \mu_+ < \mu$.

Step 3. Now, we are interested by the sign of the derivative of $\text{Re}(\lambda)$ with respect to μ at the point where λ is purely imaginary. From equation (12), we obtain:

$$\begin{aligned}
&\left(2\lambda \frac{d\lambda}{d\mu} - \lambda \frac{dB}{d\mu} - B \frac{d\lambda}{d\mu} + \frac{dC}{d\mu}\right) \left(1 + \frac{\tau}{p}\lambda\right)^p + \tau \frac{d\lambda}{d\mu} (\lambda^2 - B\lambda + C) \left(1 + \frac{\tau}{p}\lambda\right)^{p-1} \pm d \frac{d\lambda}{d\mu} = 0. \tag{18} \\
(18) \implies \frac{d\lambda}{d\mu} \left[(2\lambda - B) \left(1 + \frac{\tau}{p}\lambda\right)^p + \tau (\lambda^2 - B\lambda + C) \left(1 + \frac{\tau}{p}\lambda\right)^{p-1} \pm d \right] + \left(\frac{dC}{d\mu} - \lambda \frac{dB}{d\mu} \right) \left(1 + \frac{\tau}{p}\lambda\right)^p &= 0 \\
\implies \frac{d\lambda}{d\mu} = \frac{\left(\lambda \frac{dB}{d\mu} - \frac{dC}{d\mu} \right) \left(1 + \frac{\tau}{p}\lambda\right)^p}{(2\lambda - B) \left(1 + \frac{\tau}{p}\lambda\right)^p + \tau (\lambda^2 - B\lambda + C) \left(1 + \frac{\tau}{p}\lambda\right)^{p-1} \pm d} \\
\implies \frac{d\lambda}{d\mu} = \frac{\left(\lambda \frac{dB}{d\mu} - \frac{dC}{d\mu} \right) \left(1 + \frac{\tau}{p}\lambda\right)}{(2\lambda - B) \left(1 + \frac{\tau}{p}\lambda\right) + \tau (\lambda^2 - B\lambda + C) \pm d \left(1 + \frac{\tau}{p}\lambda\right)^{1-p}}
\end{aligned}$$

At $\lambda = i\omega$, we have $\frac{d\lambda}{d\mu} = \frac{\left(-\frac{dC}{d\mu} + i\omega\frac{dB}{d\mu}\right)\left(1 + i\frac{\tau}{p}\omega\right)}{(2i\omega - B)\left(1 + i\frac{\tau}{p}\omega\right) + \tau(C - \omega^2 - iB\omega) \pm d\left(1 + i\frac{\tau}{p}\omega\right)^{1-p}}$. Expanding give

$$\frac{d\lambda}{d\mu} = \frac{-\left(\frac{dC}{d\mu} + \frac{\tau}{p}\omega^2\frac{dB}{d\mu}\right) + i\omega\left(\frac{dB}{d\mu} - \frac{\tau}{p}\frac{dC}{d\mu}\right)}{\left(-B - 2\frac{\tau}{p}\omega^2 + \tau(C - \omega^2)\right) + i\omega\left(2 - \tau B\left(1 + \frac{1}{p}\right)\right) \pm d\left(1 + i\frac{\tau}{p}\omega\right)^{1-p}} \quad (19)$$

Writing $d\left(1 + i\frac{\tau}{p}\omega\right)^{1-p} = d\left(1 + \frac{\tau^2}{p^2}\omega^2\right)^{(1-p)/2}(\cos(1-p)\theta + i)$, the denominator of (19) becomes

$$\left(-B - 2\frac{\tau}{p}\omega^2 + \tau(C - \omega^2) \pm d\left(1 + \frac{\tau^2}{p^2}\omega^2\right)^{(1-p)/2}\cos(1-p)\theta\right) + i\left(2\omega - \omega\tau B\left(1 + \frac{1}{p}\right) \pm d\left(1 + \frac{\tau^2}{p^2}\omega^2\right)^{(1-p)/2}\sin(1-p)\theta\right) := \gamma_1 + i\gamma_2 \quad (20)$$

Mutiplied and dividing (19) by the conjuguate of (20), we obtain

$$\frac{d\lambda}{d\mu} = \frac{\left[-\gamma_1\left(\frac{dC}{d\mu} + \frac{\tau}{p}\omega^2\frac{dB}{d\mu}\right) + \omega\gamma_2\left(\frac{dB}{d\mu} - \frac{\tau}{p}\frac{dC}{d\mu}\right)\right] + i\left[\omega\gamma_1\left(\frac{dB}{d\mu} - \frac{\tau}{p}\frac{dC}{d\mu}\right) + \gamma_2\left(\frac{dC}{d\mu} + \frac{\tau}{p}\omega^2\frac{dB}{d\mu}\right)\right]}{\gamma_1^2 + \gamma_2^2}.$$

Hence,

$$\operatorname{Re}\left(\frac{d\lambda}{d\mu}\right) = \frac{-\gamma_1\left(\frac{dC}{d\mu} + \frac{\tau}{p}\omega^2\frac{dB}{d\mu}\right) + \omega\gamma_2\left(\frac{dB}{d\mu} - \frac{\tau}{p}\frac{dC}{d\mu}\right)}{\gamma_1^2 + \gamma_2^2}. \quad (21)$$

Let's compute the derivative $\frac{dB}{d\mu}$ and $\frac{dC}{d\mu}$. We have that:

$$\begin{aligned} \frac{dB}{d\mu} &= \frac{d}{d\mu} \left(\frac{C - \alpha\frac{\mu}{\varepsilon}}{1 - \mu} - d \right) = \frac{d}{d\mu} \left(\frac{\mu(1 - \alpha) - \mu^2(1 + \alpha)}{\varepsilon(1 - \mu)} - d \right) = \frac{(1 - \mu)((1 - \alpha) - 2\mu(1 + \alpha)) + \mu(1 - \alpha) - \mu^2(1 + \alpha)}{\varepsilon(1 - \mu)^2} \\ &= \frac{(1 - \alpha) - 2\mu(1 + \alpha) - \mu(1 - \alpha) + 2\mu^2(1 + \alpha) + \mu(1 - \alpha) - \mu^2(1 + \alpha)}{\varepsilon(1 - \mu)^2} = \frac{(1 - \alpha) - 2\mu(1 + \alpha) + \mu^2(1 + \alpha)}{\varepsilon(1 - \mu)^2} \\ &= \frac{1 + \alpha}{\varepsilon(1 - \mu)^2} \left(\frac{1 - \alpha}{1 + \alpha} - 2\mu + \mu^2 \right) \\ \frac{dC}{d\mu} &= \frac{d}{d\mu} \left(\frac{\mu - \mu^2(1 + \alpha)}{\varepsilon} \right) = \frac{1 - 2\mu(1 + \alpha)}{\varepsilon} \end{aligned}$$

So

$$\frac{dB}{d\mu} = \frac{1 + \alpha}{\varepsilon(1 - \mu)^2} \left(\frac{1 - \alpha}{1 + \alpha} - 2\mu + \mu^2 \right) \quad \text{and} \quad \frac{dC}{d\mu} = \frac{1 - 2\mu(1 + \alpha)}{\varepsilon} \quad (22)$$

We have that, at $\tau = 0$, $\theta = 0$, $\gamma_1 = -B \pm d$, $\gamma_2 = 2\omega$ and (21) becomes

$$\operatorname{Re}\left(\frac{d\lambda}{d\mu}\right) \Big|_{\tau=0} = \frac{(B \mp d)\frac{dC}{d\mu} + 2\omega^2\frac{dB}{d\mu}}{(-B \pm d)^2 + 4\omega^2} = \frac{2\omega^2\frac{dB}{d\mu}}{4\omega^2} = \frac{1}{2}\frac{dB}{d\mu} \quad (23)$$

since from equation (14) we have, $-B \pm d = 0$. We also have from equation (14) that $\mu = \frac{1 - \alpha}{1 + \alpha} := \mu_1$, $\mu = \frac{2\varepsilon d + 1 - \alpha \pm \sqrt{(2\varepsilon d - (1 + 3\alpha))^2 + 4\varepsilon(1 + \alpha)^2}}{2\varepsilon(1 + \alpha)} := \mu_{2,3}$, provided that $2\varepsilon d \leq 1 + 3\alpha - \sqrt{8\alpha + 8\alpha^2}$.

Next, we compute $\operatorname{Re}\left(\frac{d\lambda}{d\mu}\right) \Big|_{\tau=0}$ defined in (23) for these values of μ .

- For $\mu = \frac{1-\alpha}{1+\alpha} := \mu_1$, we have:

$$\begin{aligned} \operatorname{Re} \left(\frac{d\lambda}{d\mu} \right) \Big|_{\tau=0} &= \frac{1+\alpha}{2\varepsilon \left(1 - \frac{1-\alpha}{1+\alpha} \right)^2} \left(\frac{1-\alpha}{1+\alpha} - 2 \left(\frac{1-\alpha}{1+\alpha} \right) + \left(\frac{1-\alpha}{1+\alpha} \right)^2 \right) = \frac{1+\alpha}{2\varepsilon \left(1 - \frac{1-\alpha}{1+\alpha} \right)^2} \left(\left(\frac{1-\alpha}{1+\alpha} \right)^2 - \frac{1-\alpha}{1+\alpha} \right) \\ &= -\frac{(1+\alpha) \left(\frac{1-\alpha}{1+\alpha} \right)}{2\varepsilon \left(1 - \frac{1-\alpha}{1+\alpha} \right)^2} \left(1 - \frac{1-\alpha}{1+\alpha} \right) = -\frac{1-\alpha}{2\varepsilon \left(1 - \frac{1-\alpha}{1+\alpha} \right)} = -\frac{1-\alpha}{2\varepsilon \left(\frac{2\alpha}{1+\alpha} \right)} = -\frac{(1-\alpha)(1+\alpha)}{4\varepsilon\alpha} < 0. \end{aligned}$$

- For $\mu = \frac{2\varepsilon d + 1 - \alpha \pm \sqrt{(2\varepsilon d - x_-)(2\varepsilon d - x_+)}}{2(1+\alpha)} := \mu_-, \mu_+$, with $2\varepsilon d \leq x_-$, we have

$$\begin{aligned} \operatorname{Re} \left(\frac{d\lambda}{d\mu} \right) \Big|_{\tau=0} &= \frac{1+\alpha}{2\varepsilon(1-\mu_{\pm})^2} \left(\frac{1-\alpha}{1+\alpha} - 2\mu_{\pm} + \mu_{\pm}^2 \right) = \frac{1+\alpha}{2\varepsilon(1-\mu_{\pm})^2} \left(\frac{1-\alpha}{1+\alpha} + (1-\mu_{\pm})^2 - 1 \right) \\ &= \frac{1+\alpha}{2\varepsilon(1-\mu_{\pm})^2} \left((1-\mu_{\pm})^2 - \frac{2\alpha}{1+\alpha} \right) \end{aligned}$$

We have that: $1 - \mu_{\pm} = 1 - \frac{2\varepsilon d + 1 - \alpha \pm \sqrt{(2\varepsilon d - x_-)(2\varepsilon d - x_+)}}{2(1+\alpha)} = \frac{1+3\alpha - 2\varepsilon d \mp \sqrt{(2\varepsilon d - x_-)(2\varepsilon d - x_+)}}{2(1+\alpha)}$. Hence,

$$\begin{aligned} (1 - \mu_{\pm})^2 - \frac{2\alpha}{1+\alpha} &= \frac{\left(1 + 3\alpha - 2\varepsilon d \mp \sqrt{(2\varepsilon d - x_-)(2\varepsilon d - x_+)} \right)^2}{4(1+\alpha)^2} - \frac{2\alpha}{1+\alpha} \\ &= \frac{\left(1 + 3\alpha - 2\varepsilon d \mp \sqrt{(2\varepsilon d - x_-)(2\varepsilon d - x_+)} \right)^2 - 8\alpha(1+\alpha)}{4(1+\alpha)^2} \\ &= \frac{\left(1 + 3\alpha - 2\varepsilon d \mp \sqrt{(2\varepsilon d - x_-)(2\varepsilon d - x_+)} - \sqrt{8\alpha(1+\alpha)} \right) \left(1 + 3\alpha - 2\varepsilon d \mp \sqrt{(2\varepsilon d - x_-)(2\varepsilon d - x_+)} + \sqrt{8\alpha(1+\alpha)} \right)}{4(1+\alpha)^2} \\ &= \frac{\left((x_- - 2\varepsilon d) \mp \sqrt{(2\varepsilon d - x_-)(2\varepsilon d - x_+)} \right) \left((x_+ - 2\varepsilon d) \mp \sqrt{(2\varepsilon d - x_-)(2\varepsilon d - x_+)} \right)}{4(1+\alpha)^2} \end{aligned}$$

– If $2\varepsilon d = x_-$ then $\mu_- = \mu_+$ and $(1 - \mu_{\pm})^2 - \frac{2\alpha}{1+\alpha} = 0$.

– For $\mu = \mu_-$, with $2\varepsilon d < x_- < x_+$, we have that

$$(1 - \mu_-)^2 - \frac{2\alpha}{1+\alpha} = \frac{\left((x_- - 2\varepsilon d) + \sqrt{(2\varepsilon d - x_-)(2\varepsilon d - x_+)} \right) \left((x_+ - 2\varepsilon d) + \sqrt{(2\varepsilon d - x_-)(2\varepsilon d - x_+)} \right)}{4(1+\alpha)^2} > 0$$

– For $\mu = \mu_+$, with $2\varepsilon d < x_- < x_+$, we have that

$$\begin{aligned} (1 - \mu_+)^2 - \frac{2\alpha}{1+\alpha} &= \frac{\left((x_- - 2\varepsilon d) - \sqrt{(x_- - 2\varepsilon d)(x_+ - 2\varepsilon d)} \right) \left((x_+ - 2\varepsilon d) - \sqrt{(x_- - 2\varepsilon d)(x_+ - 2\varepsilon d)} \right)}{4(1+\alpha)^2} \\ &= \frac{2(x_- - 2\varepsilon d)(x_+ - 2\varepsilon d) - (x_- - 2\varepsilon d + x_+ - 2\varepsilon d) \sqrt{(x_- - 2\varepsilon d)(x_+ - 2\varepsilon d)}}{4(1+\alpha)^2} \\ &= \frac{\sqrt{(x_- - 2\varepsilon d)(x_+ - 2\varepsilon d)} \left(2\sqrt{(x_- - 2\varepsilon d)(x_+ - 2\varepsilon d)} - (x_- - 2\varepsilon d + x_+ - 2\varepsilon d) \right)}{4(1+\alpha)^2} \\ &= -\frac{\sqrt{(x_- - 2\varepsilon d)(x_+ - 2\varepsilon d)} \left((\sqrt{x_- - 2\varepsilon d})^2 + (\sqrt{x_+ - 2\varepsilon d})^2 - 2\sqrt{x_- - 2\varepsilon d}\sqrt{x_+ - 2\varepsilon d} \right)}{4(1+\alpha)^2} \\ &= -\frac{\sqrt{(x_- - 2\varepsilon d)(x_+ - 2\varepsilon d)} \left(\sqrt{x_- - 2\varepsilon d} - \sqrt{x_+ - 2\varepsilon d} \right)^2}{4(1+\alpha)^2} < 0 \end{aligned}$$

We conclude that

$$\operatorname{Re} \left(\frac{d\lambda}{d\mu} \right) \Big|_{\tau=0} \begin{cases} = 0 & \text{if } 2\varepsilon d = x_- \\ < 0 & \text{if } \mu = \mu_+ \text{ and } 2\varepsilon d < x_- \\ > 0 & \text{if } \mu = \mu_- \text{ and } 2\varepsilon d < x_- \end{cases}$$

Step 4. Next we apply the implicit function theorem to equation (13). We want to determine (μ, ω) as a function of τ near $\tau = 0$ for the points (μ_1, ω_1) , (μ_-, ω_-) and (μ_+, ω_+) .

If p is a positive integer, we have that

$$\left(1 + i \frac{\tau}{p} w \right)^p = \sum_{k=0}^p \binom{p}{k} i^k \left(\frac{\tau}{p} w \right)^k = \sum_{k=0}^{[p/2]} (-1)^k \binom{p}{2k} \left(\frac{\tau}{p} w \right)^{2k} + i \sum_{k=0}^{[(p-1)/2]} (-1)^k \binom{p}{2k+1} \left(\frac{\tau}{p} w \right)^{2k+1} := S_R(\tau, \omega, p) + i S_I(\tau, \omega, p),$$

where $k < p$ is a positive integer. We have $S_R(0, \omega, p) = 1$ and $S_I(0, \omega, p) = 0$. We recall (13):

$$\begin{aligned} & \left[(C - \omega^2 - i\omega B) \left(1 + i \frac{\tau}{p} w \right)^p - id\omega \right] \left[(C - \omega^2 - i\omega B) \left(1 + i \frac{\tau}{p} w \right)^p + id\omega \right] = 0. \\ \implies & ((C - \omega^2 - i\omega B) (S_R + iS_I) - id\omega) ((C - \omega^2 - i\omega B) (S_R + iS_I) + id\omega) = 0. \\ \implies & (S_R (C - \omega^2) + \omega B S_I - i\omega B S_R + iS_I (C - \omega^2) - id\omega) (S_R (C - \omega^2) + \omega B S_I - i\omega B S_R + iS_I (C - \omega^2) + id\omega) = 0. \\ \implies & [(S_R (C - \omega^2) + \omega B S_I) + i (S_I (C - \omega^2) - \omega B S_R - d\omega)] [(S_R (C - \omega^2) + \omega B S_I) + i (S_I (C - \omega^2) - \omega B S_R + d\omega)] = 0. \\ \implies & [(S_R (C - \omega^2) + \omega B S_I)^2 - (S_I (C - \omega^2) - \omega B S_R - d\omega) (S_I (C - \omega^2) - \omega B S_R + d\omega)] \\ & + i [(S_R (C - \omega^2) + \omega B S_I) (S_I (C - \omega^2) - \omega B S_R - d\omega) + (S_R (C - \omega^2) + \omega B S_I) (S_I (C - \omega^2) - \omega B S_R + d\omega)] = 0. \\ \implies & [(S_R (C - \omega^2) + \omega B S_I)^2 - (S_I (C - \omega^2) - \omega B S_R)^2 + d^2 \omega^2] + 2i [(S_R (C - \omega^2) + \omega B S_I) (S_I (C - \omega^2) - \omega B S_R)] = 0. \end{aligned}$$

- For (μ_1, ω_1) :

$$\text{Let } F : \mathbb{R}^3 \mapsto \mathbb{R}^2 \text{ such that } F(\tau, \mu, \omega) = \begin{pmatrix} (S_R (C - \omega^2) + \omega B S_I)^2 - (S_I (C - \omega^2) - \omega B S_R)^2 + d^2 \omega^2 \\ 2 (S_R (C - \omega^2) + \omega B S_I) (S_I (C - \omega^2) - \omega B S_R) \end{pmatrix} := \begin{pmatrix} F_1(\tau, \mu, \omega) \\ F_2(\tau, \mu, \omega) \end{pmatrix}$$

Next, we show that $\det \begin{pmatrix} \frac{\partial F_1}{\partial \mu} & \frac{\partial F_1}{\partial \omega} \\ \frac{\partial F_2}{\partial \mu} & \frac{\partial F_2}{\partial \omega} \end{pmatrix} \neq 0$. We start by finding the expression of each derivative:

$$\frac{\partial F_1}{\partial \mu} = 2 \left(S_R \frac{dC}{d\mu} + \omega S_I \frac{dB}{d\mu} \right) (S_R (C - \omega^2) + \omega B S_I) - 2 \left(S_I \frac{dC}{d\mu} - \omega S_R \frac{dB}{d\mu} \right) (S_I (C - \omega^2) - \omega B S_R)$$

$$\begin{aligned} \frac{\partial F_1}{\partial \omega} &= 2 \left((C - \omega^2) \frac{\partial S_R}{\partial \omega} - 2\omega S_R + \omega B \frac{\partial S_I}{\partial \omega} + B S_I \right) (S_R (C - \omega^2) + \omega B S_I) \\ &\quad - 2 \left((C - \omega^2) \frac{\partial S_I}{\partial \omega} - 2\omega S_I - \omega B \frac{\partial S_R}{\partial \omega} - B S_R \right) (S_I (C - \omega^2) - \omega B S_R) + 2d^2 \omega \end{aligned}$$

$$\frac{\partial F_2}{\partial \mu} = \left(S_R \frac{dC}{d\mu} + \omega S_I \frac{dB}{d\mu} \right) (S_I (C - \omega^2) - \omega B S_R) + 2 \left(S_I \frac{dC}{d\mu} - \omega S_R \frac{dB}{d\mu} \right) (S_R (C - \omega^2) + \omega B S_I)$$

$$\begin{aligned} \frac{\partial F_2}{\partial \omega} &= 2 \left((C - \omega^2) \frac{\partial S_R}{\partial \omega} - 2\omega S_R + \omega B \frac{\partial S_I}{\partial \omega} + B S_I \right) (S_I (C - \omega^2) - \omega B S_R) \\ &\quad + 2 \left((C - \omega^2) \frac{\partial S_I}{\partial \omega} - 2\omega S_I - \omega B \frac{\partial S_R}{\partial \omega} - B S_R \right) (S_R (C - \omega^2) + \omega B S_I) \end{aligned}$$

with $\frac{\partial S_R}{\partial \omega} = \sum_{k=1}^{[p/2]} (-1)^k \binom{p}{2k} (2k) \left(\frac{\tau}{p}\right) \left(\frac{\tau}{p} w\right)^{2k-1}$ and $\frac{\partial S_I}{\partial \omega} = \sum_{k=0}^{[(p-1)/2]} (-1)^k \binom{p}{2k+1} (2k+1) \left(\frac{\tau}{p}\right) \left(\frac{\tau}{p} w\right)^{2k}$.

At $x = x^*$, $C - \omega^2 = 0$, $B = -d$, $S_R = 1$, $S_I = 0$, $\frac{dC}{d\mu} = \frac{2\alpha - 1}{\varepsilon}$, $\frac{dB}{d\mu} = -\frac{(1-\alpha)(1+\alpha)}{2\varepsilon\alpha}$, $\frac{\partial S_R}{\partial \omega} = 0$ and $\frac{\partial S_I}{\partial \omega} = 0$.

Hence,

$$\begin{pmatrix} \frac{\partial F_1}{\partial \mu} & \frac{\partial F_1}{\partial \omega} \\ \frac{\partial F_2}{\partial \mu} & \frac{\partial F_2}{\partial \omega} \end{pmatrix} \Big|_{x=x^*} = \begin{pmatrix} -2\omega^2 B \frac{dB}{d\mu} & -2\omega B + 2d^2 \omega \\ -2\omega B \frac{dC}{d\mu} & 4\omega^2 B \end{pmatrix} \Big|_{x=x^*} = \begin{pmatrix} \omega^2 d \frac{(1-\alpha)(1+\alpha)}{\varepsilon\alpha} & 2\omega d(1+d) \\ 2\omega d \frac{2\alpha-1}{\varepsilon} & -4\omega^2 d \end{pmatrix}.$$

Therefore, $\det \begin{pmatrix} \frac{\partial F_1}{\partial \mu} & \frac{\partial F_1}{\partial \omega} \\ \frac{\partial F_2}{\partial \mu} & \frac{\partial F_2}{\partial \omega} \end{pmatrix} \Big|_{x=x^*} = -4\omega^4 d^2 \frac{(1-\alpha)(1+\alpha)}{\varepsilon\alpha} - 4\omega^2 d^2 (1+d) \frac{2\alpha-1}{\varepsilon}$
 $= -4 \frac{\omega^2 d^2}{\varepsilon} \left(\frac{\alpha(1-\alpha)}{\varepsilon(1+\alpha)} \frac{(1-\alpha)(1+\alpha)}{\alpha} - (1+d)(2\alpha-1) \right) = -4 \frac{\omega^2 d^2}{\varepsilon} \left(\frac{(1-\alpha)^2}{\varepsilon} - (1+d)(2\alpha-1) \right).$

This determinant is zero if and only if $d = \frac{(1-\alpha)^2}{\varepsilon(2\alpha-1)} - 1$ and nonzero otherwise.

So the theorem guarantees that the equations implicitly determine (μ, ω) as a function of τ .

Next, we differentiate F_1 and F_2 with respect to τ . We obtain: $\frac{\partial F_1}{\partial \tau} = (C - \omega^2) \frac{\partial S_R}{\partial \tau} + \omega B \frac{\partial S_I}{\partial \tau}$ and $\frac{\partial F_2}{\partial \tau} = (C - \omega^2) \frac{\partial S_I}{\partial \tau} - \omega B \frac{\partial S_R}{\partial \tau}$, with $\frac{\partial S_R}{\partial \tau} = \sum_{k=0}^{[p/2]} (-1)^k \binom{p}{2k} (2k) \left(\frac{\omega}{p}\right) \left(\frac{\tau}{p} w\right)^{2k-1}$ and $\frac{\partial S_I}{\partial \tau} = \sum_{k=0}^{[(p-1)/2]} (-1)^k \binom{p}{2k+1} (2k+1) \left(\frac{\omega}{p}\right) \left(\frac{\tau}{p} w\right)^{2k}$.

We then have that:

$$\begin{pmatrix} \frac{\partial F_1}{\partial \tau} \\ \frac{\partial F_2}{\partial \tau} \end{pmatrix} \Big|_{x=x^*} = \begin{pmatrix} (C - \omega^2) \frac{\partial S_R}{\partial \tau} + \omega B \frac{\partial S_I}{\partial \tau} \\ (C - \omega^2) \frac{\partial S_I}{\partial \tau} - \omega B \frac{\partial S_R}{\partial \tau} \end{pmatrix} \Big|_{x=x^*} = \begin{pmatrix} \omega^2 B \\ (C - \omega^2) \omega \end{pmatrix} = \begin{pmatrix} -\frac{\alpha(1-\alpha)}{\varepsilon(1+\alpha)} d \\ 0 \end{pmatrix}$$

Then the differential of (μ, ω) with respect to τ at x^* is

$$\begin{aligned} \frac{d}{d\tau}(\mu, \omega) &= - \left[\begin{pmatrix} \frac{\partial F_1}{\partial \mu} & \frac{\partial F_1}{\partial \omega} \\ \frac{\partial F_2}{\partial \mu} & \frac{\partial F_2}{\partial \omega} \end{pmatrix} \Big|_{x=x^*} \right]^{-1} \times \begin{pmatrix} \frac{\partial F_1}{\partial \tau} \\ \frac{\partial F_2}{\partial \tau} \end{pmatrix} \Big|_{x=x^*} \\ &= - \frac{\varepsilon^2}{(1-\alpha)^2} \begin{pmatrix} 0 & 2\sqrt{\frac{\alpha(1-\alpha)}{\varepsilon(1+\alpha)}} \\ -\frac{(1-\alpha)(1+\alpha)}{2\varepsilon\alpha} \sqrt{\frac{\alpha(1-\alpha)}{\varepsilon(1+\alpha)}} & \frac{2\alpha-1}{\varepsilon} \end{pmatrix} \times \begin{pmatrix} -\frac{\alpha(1-\alpha)}{\varepsilon(1+\alpha)} d \\ 0 \end{pmatrix} \\ &= - \frac{\varepsilon^2}{(1-\alpha)^2} \begin{pmatrix} 0 \\ \frac{(1-\alpha)(1+\alpha)}{2\varepsilon\alpha} \sqrt{\frac{\alpha(1-\alpha)}{\varepsilon(1+\alpha)}} \frac{\alpha(1-\alpha)}{\varepsilon(1+\alpha)} d \end{pmatrix} = \begin{pmatrix} 0 \\ -\frac{d}{2} \sqrt{\frac{\alpha(1-\alpha)}{\varepsilon(1+\alpha)}} \end{pmatrix} \end{aligned}$$

Hence, $\frac{d\mu}{d\tau} \Big|_{\mu=\mu_1} = 0$ and $\frac{d\omega}{d\tau} \Big|_{\omega=\omega_1} < 0$.

- For (μ_-, ω_-) and (μ_+, ω_+) : Equation (13) with the plus sign becomes

$$(S_R(C - \omega^2) + \omega BS_I) + i(S_I(C - \omega^2) - \omega BS_R + d\omega) = 0. \quad (24)$$

Let $F : \mathbb{R}^3 \mapsto \mathbb{R}^2$ such that $F(\tau, \mu, \omega) = \begin{pmatrix} S_R(C - \omega^2) + \omega BS_I \\ S_I(C - \omega^2) - \omega BS_R + d\omega \end{pmatrix} := \begin{pmatrix} F_1(\tau, \mu, \omega) \\ F_2(\tau, \mu, \omega) \end{pmatrix}$ and satisfies $F(x_-^*) = 0 = F(x_+^*)$, where $x_-^* = (0, \mu_-, \omega_-)$ and $x_+^* = (0, \mu_+, \omega_+)$. We want to solve $F(\tau, \mu, \omega) = 0$ for (μ, ω) as a function of τ .

Next, we show that $\det \begin{pmatrix} \frac{\partial F_1}{\partial \mu} & \frac{\partial F_1}{\partial \omega} \\ \frac{\partial F_2}{\partial \mu} & \frac{\partial F_2}{\partial \omega} \end{pmatrix} \neq 0$. We start by finding the expression of each derivative:

$$\begin{aligned} \frac{\partial F_1}{\partial \mu} &= S_R \frac{dC}{d\mu} + \omega S_I \frac{dB}{d\mu}, \quad \frac{\partial F_1}{\partial \omega} = (C - \omega^2) \frac{\partial S_R}{\partial \omega} - 2\omega S_R + \omega B \frac{\partial S_I}{\partial \omega} + BS_I, \quad \frac{\partial F_2}{\partial \mu} = S_I \frac{dC}{d\mu} - \omega S_R \frac{dB}{d\mu}, \quad \frac{\partial F_2}{\partial \omega} = (C - \omega^2) \frac{\partial S_I}{\partial \omega} - \\ &2\omega S_I - \omega B \frac{\partial S_R}{\partial \omega} - BS_R + d, \quad \text{with } \frac{\partial S_R}{\partial \omega} = \sum_{k=1}^{[p/2]} (-1)^k \binom{p}{2k} (2k) \left(\frac{\tau}{p}\right) \left(\frac{\tau}{p} w\right)^{2k-1} \quad \text{and} \quad \frac{\partial S_I}{\partial \omega} = \sum_{k=0}^{[(p-1)/2]} (-1)^k \binom{p}{2k+1} (2k+1) \left(\frac{\tau}{p}\right) \left(\frac{\tau}{p} w\right)^{2k}. \end{aligned}$$

Hence,

$$\begin{pmatrix} \frac{\partial F_1}{\partial \mu} & \frac{\partial F_1}{\partial \omega} \\ \frac{\partial F_2}{\partial \mu} & \frac{\partial F_2}{\partial \omega} \end{pmatrix} = \begin{pmatrix} S_R \frac{dC}{d\mu} + \omega S_I \frac{dB}{d\mu} & (C - \omega^2) \frac{\partial S_R}{\partial \omega} - 2\omega S_R + \omega B \frac{\partial S_I}{\partial \omega} + BS_I \\ S_I \frac{dC}{d\mu} - \omega S_R \frac{dB}{d\mu} & (C - \omega^2) \frac{\partial S_I}{\partial \omega} - 2\omega S_I - \omega B \frac{\partial S_R}{\partial \omega} - BS_R + d \end{pmatrix}.$$

At $x = x_-^*$, $C - \omega_-^2 = 0$, $B = d$, $S_R = 1$, $S_I = 0$, $\frac{dC}{d\mu} = \frac{1 - 2\mu_-(1 + \alpha)}{\varepsilon}$, $\frac{dB}{d\mu} = \frac{1 + \alpha}{\varepsilon(1 - \mu_-)^2} \left(\frac{1 - \alpha}{1 + \alpha} - 2\mu_- + \mu_-^2 \right)$, $\frac{\partial S_R}{\partial \omega} = 0$ and $\frac{\partial S_I}{\partial \omega} = 0$.

At $x = x_+^*$, $C - \omega_+^2 = 0$, $B = d$, $S_R = 1$, $S_I = 0$, $\frac{dC}{d\mu} = \frac{1 - 2\mu_+(1 + \alpha)}{\varepsilon}$, $\frac{dB}{d\mu} = \frac{1 + \alpha}{\varepsilon(1 - \mu_+)^2} \left(\frac{1 - \alpha}{1 + \alpha} - 2\mu_+ + \mu_+^2 \right)$, $\frac{\partial S_R}{\partial \omega} = 0$ and $\frac{\partial S_I}{\partial \omega} = 0$.

Hence,

$$\begin{pmatrix} \frac{\partial F_1}{\partial \mu} & \frac{\partial F_1}{\partial \omega} \\ \frac{\partial F_2}{\partial \mu} & \frac{\partial F_2}{\partial \omega} \end{pmatrix} \Big|_{x=x_-^*} = \begin{pmatrix} \frac{1 - 2\mu_-(1 + \alpha)}{\varepsilon} & -2\sqrt{\frac{2\varepsilon d + (\alpha - 2\varepsilon d)\mu_-}{\varepsilon}} \\ -\frac{1 + \alpha}{\varepsilon(1 - \mu_-)^2} \left(\frac{1 - \alpha}{1 + \alpha} - 2\mu_- + \mu_-^2 \right) \sqrt{\frac{2\varepsilon d + (\alpha - 2\varepsilon d)\mu_-}{\varepsilon}} & 0 \end{pmatrix}$$

and

$$\begin{pmatrix} \frac{\partial F_1}{\partial \mu} & \frac{\partial F_1}{\partial \omega} \\ \frac{\partial F_2}{\partial \mu} & \frac{\partial F_2}{\partial \omega} \end{pmatrix} \Big|_{x=x_+^*} = \begin{pmatrix} \frac{1 - 2\mu_+(1 + \alpha)}{\varepsilon} & -2\sqrt{\frac{2\varepsilon d + (\alpha - 2\varepsilon d)\mu_+}{\varepsilon}} \\ -\frac{1 + \alpha}{\varepsilon(1 - \mu_+)^2} \left(\frac{1 - \alpha}{1 + \alpha} - 2\mu_+ + \mu_+^2 \right) \sqrt{\frac{2\varepsilon d + (\alpha - 2\varepsilon d)\mu_+}{\varepsilon}} & 0 \end{pmatrix}.$$

$$\begin{aligned} - \text{Therefore, } \det \begin{pmatrix} \frac{\partial F_1}{\partial \mu} & \frac{\partial F_1}{\partial \omega} \\ \frac{\partial F_2}{\partial \mu} & \frac{\partial F_2}{\partial \omega} \end{pmatrix} \Big|_{x=x_-^*} &= -2 \left(\frac{(1 - \alpha) - 2\mu(1 + \alpha) + \mu^2(1 + \alpha)}{\varepsilon(1 - \mu)^2} \right) \left(\frac{2\varepsilon d + (\alpha - 2\varepsilon d)\mu_-}{\varepsilon} \right) \\ &= -2 \left(\frac{(1 - \alpha) + (\mu^2 - 2\mu)(1 + \alpha)}{\varepsilon^2(1 - \mu)^2} \right) (2(1 - \mu_-)\varepsilon d + \alpha) = -2 \left(\frac{(1 - \alpha) + (1 + \alpha)(1 - \mu_-)^2 - (1 + \alpha)}{\varepsilon^2(1 - \mu)^2} \right) (2(1 - \mu_-)\varepsilon d + \alpha\mu_-) \end{aligned}$$

$= -2 \left(\frac{(1+\alpha)(1-\mu_-)^2 - 2\alpha}{\varepsilon^2(1-\mu)^2} \right) (2(1-\mu_-)\varepsilon d + \alpha\mu_-)$. We have shown above (see bottom of page 16) that $(1-\mu_-)^2 - \frac{2\alpha}{(1+\alpha)} > 0$ and $(1-\mu_+)^2 - \frac{2\alpha}{(1+\alpha)} < 0$. Therefore, this determinant is zero if and only if $2(1-\mu_-)\varepsilon d + \alpha\mu_- = 0$; otherwise, it is not zero and the theorem guarantees that solution exists.

Next, we differentiate F_1 and F_2 with respect to τ . We obtain: $\frac{\partial F_1}{\partial \tau} = (C - \omega^2) \frac{\partial S_R}{\partial \tau} + \omega B \frac{\partial S_I}{\partial \tau}$ and $\frac{\partial F_2}{\partial \tau} = (C - \omega^2) \frac{\partial S_I}{\partial \tau} - \omega B \frac{\partial S_R}{\partial \tau}$, with $\frac{\partial S_R}{\partial \tau} = \sum_{k=1}^{[p/2]} (-1)^k \binom{p}{2k} (2k) \left(\frac{\omega}{p} \right) \left(\frac{\tau}{p} w \right)^{2k-1}$ and $\frac{\partial S_I}{\partial \tau} = \sum_{k=0}^{[(p-1)/2]} (-1)^k \binom{p}{2k+1} (2k+1) \left(\frac{\omega}{p} \right) \left(\frac{\tau}{p} w \right)^{2k}$.

We then have that:

$$\begin{pmatrix} \frac{\partial F_1}{\partial \tau} \\ \frac{\partial F_2}{\partial \tau} \end{pmatrix} \Big|_{x=x_-^*} = \begin{pmatrix} (C - \omega^2) \frac{\partial S_R}{\partial \omega} + \omega B \frac{\partial S_I}{\partial \tau} \\ (C - \omega^2) \frac{\partial S_I}{\partial \omega} - \omega B \frac{\partial S_R}{\partial \tau} \end{pmatrix} \Big|_{x=x_-^*} = \begin{pmatrix} \omega_-^2 B \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{2\varepsilon d + (\alpha - 2\varepsilon d)\mu_-}{\varepsilon} d \\ 0 \end{pmatrix}$$

Then the differential of (μ, ω) with respect to τ at x_-^* is

$$\begin{aligned} \frac{d}{d\tau}(\mu, \omega) &= - \left[\begin{pmatrix} \frac{\partial F_1}{\partial \mu} & \frac{\partial F_1}{\partial \omega} \\ \frac{\partial F_2}{\partial \mu} & \frac{\partial F_2}{\partial \omega} \end{pmatrix} \Big|_{x=x_-^*} \right]^{-1} \times \begin{pmatrix} \frac{\partial F_1}{\partial \tau} \\ \frac{\partial F_2}{\partial \tau} \end{pmatrix} \Big|_{x=x_-^*} \\ &= -\frac{1}{\det} \begin{pmatrix} 0 & 2\sqrt{\frac{2\varepsilon d + (\alpha - 2\varepsilon d)\mu_-}{\varepsilon}} \\ \frac{1+\alpha}{\varepsilon(1-\mu_-)^2} \left(\frac{1-\alpha}{1+\alpha} - 2\mu_- + \mu_-^2 \right) \sqrt{\frac{2\varepsilon d + (\alpha - 2\varepsilon d)\mu_-}{\varepsilon}} & \frac{1-2\mu_-(1+\alpha)}{\varepsilon} \end{pmatrix} \times \begin{pmatrix} \frac{2\varepsilon d + (\alpha - 2\varepsilon d)\mu_-}{\varepsilon} \\ 0 \end{pmatrix} \\ &= -\frac{1}{\det} \begin{pmatrix} 0 \\ \frac{1+\alpha}{\varepsilon(1-\mu_-)^2} \left(\frac{1-\alpha}{1+\alpha} - 2\mu_- + \mu_-^2 \right) \left(\frac{2\varepsilon d + (\alpha - 2\varepsilon d)\mu_-}{\varepsilon} \right) \sqrt{\frac{2\varepsilon d + (\alpha - 2\varepsilon d)\mu_-}{\varepsilon}} \end{pmatrix} \end{aligned}$$

Hence, $\frac{d\mu}{d\tau} \Big|_{\mu=\mu_-} = 0$ and $\frac{d\omega}{d\tau} \Big|_{\omega=\omega_-} < 0$.

– Therefore, $\det \begin{pmatrix} \frac{\partial F_1}{\partial \mu} & \frac{\partial F_1}{\partial \omega} \\ \frac{\partial F_2}{\partial \mu} & \frac{\partial F_2}{\partial \omega} \end{pmatrix} \Big|_{x=x_+^*} = - \left(\frac{(1-\alpha) - 2\mu(1+\alpha) + \mu^2(1+\alpha)}{\varepsilon(1-\mu)^2} \right) \left(\frac{2\varepsilon d + (\alpha - 2\varepsilon d)\mu_+}{\varepsilon} \right) \neq 0$. So the

theorem guarantees that solution exists.

Next, we differentiate F_1 and F_2 with respect to τ . We obtain: $\frac{\partial F_1}{\partial \tau} = (C - \omega^2) \frac{\partial S_R}{\partial \tau} + \omega B \frac{\partial S_I}{\partial \tau}$ and $\frac{\partial F_2}{\partial \tau} = (C - \omega^2) \frac{\partial S_I}{\partial \tau} - \omega B \frac{\partial S_R}{\partial \tau}$, with $\frac{\partial S_R}{\partial \tau} = \sum_{k=1}^{[p/2]} (-1)^k \binom{p}{2k} (2k) \left(\frac{\omega}{p} \right) \left(\frac{\tau}{p} w \right)^{2k-1}$ and $\frac{\partial S_I}{\partial \tau} = \sum_{k=0}^{[(p-1)/2]} (-1)^k \binom{p}{2k+1} (2k+1) \left(\frac{\omega}{p} \right) \left(\frac{\tau}{p} w \right)^{2k}$.

We then have that:

$$\begin{pmatrix} \frac{\partial F_1}{\partial \tau} \\ \frac{\partial F_2}{\partial \tau} \end{pmatrix} \Big|_{x=x_+^*} = \begin{pmatrix} (C - \omega^2) \frac{\partial S_R}{\partial \omega} + \omega B \frac{\partial S_I}{\partial \tau} \\ (C - \omega^2) \frac{\partial S_I}{\partial \omega} - \omega B \frac{\partial S_R}{\partial \tau} \end{pmatrix} \Big|_{x=x_+^*} = \begin{pmatrix} \omega_+^2 B \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{2\varepsilon d + (\alpha - 2\varepsilon d)\mu_+}{\varepsilon} d \\ 0 \end{pmatrix}$$

Then the differential of (μ, ω) with respect to τ at x_+^* is

$$\begin{aligned}
\frac{d}{d\tau}(\mu, \omega) &= - \left[\begin{pmatrix} \frac{\partial F_1}{\partial \mu} & \frac{\partial F_1}{\partial \omega} \\ \frac{\partial F_2}{\partial \mu} & \frac{\partial F_2}{\partial \omega} \end{pmatrix} \Big|_{x=x_+^*} \right]^{-1} \times \begin{pmatrix} \frac{\partial F_1}{\partial \tau} \\ \frac{\partial F_2}{\partial \tau} \end{pmatrix} \Big|_{x=x_+^*} \\
&= -\frac{1}{det} \begin{pmatrix} 0 & 2\sqrt{\frac{2\varepsilon d + (\alpha - 2\varepsilon d)\mu_+}{\varepsilon}} \\ \frac{1+\alpha}{\varepsilon(1-\mu_+)^2} \left(\frac{1-\alpha}{1+\alpha} - 2\mu_+ + \mu_+^2 \right) \sqrt{\frac{2\varepsilon d + (\alpha - 2\varepsilon d)\mu_+}{\varepsilon}} & \frac{1-2\mu_+(1+\alpha)}{\varepsilon} \end{pmatrix} \times \begin{pmatrix} \frac{2\varepsilon d + (\alpha - 2\varepsilon d)\mu_+}{\varepsilon} \\ 0 \end{pmatrix} \\
&= -\frac{1}{det} \begin{pmatrix} 0 \\ \frac{1+\alpha}{\varepsilon(1-\mu_+)^2} \left(\frac{1-\alpha}{1+\alpha} - 2\mu_+ + \mu_+^2 \right) \left(\frac{2\varepsilon d + (\alpha - 2\varepsilon d)\mu_+}{\varepsilon} \right) \sqrt{\frac{2\varepsilon d + (\alpha - 2\varepsilon d)\mu_+}{\varepsilon}} \end{pmatrix} \\
\text{Hence, } \frac{d\mu}{d\tau} \Big|_{\mu=\mu_+} &= 0 \text{ and } \frac{d\omega}{d\tau} \Big|_{\omega=\omega_+} < 0.
\end{aligned}$$

3.5 Numerical analysis

We recall the expression of the characteristic equation (12)

$$f_{\pm}(\lambda) := (\lambda^2 - B\lambda + C) \left(1 + \frac{\tau}{p}\lambda \right)^p \pm d\lambda = 0. \quad (25)$$

p	(μ_1, ω_1)	(μ_2, ω_2)	(μ_3, ω_3)
0	(0.0869, 0.7080)	(0.2228, 0.9861)	(0.2903, 1.0317)
1	(0.0420, 0.5285)	(0.2621, 1.0280)	(0.2644, 1.0105)
2	(0.0337, 0.4819)	(0.2621, 1.0280)	(0.2663, 1.0255)
2	(0.0293, 0.4558)	(0.2621, 1.0280)	(0.2663, 1.0255)

Table 1: Values of Hopf bifurcation points μ and ω for $\varepsilon = 0.15$, $\alpha = 0.55$, $d = 0.1$ and $\tau = 5$

τ	(μ_1, ω_1)	(μ_2, ω_2)	(μ_3, ω_3)	
0.1	(0.0867, 0.7109)	(0.2231, 0.9914)	(0.2902, 1.0266)	
0.2	(0.0861, 0.7124)	(0.2241, 0.9974)	(0.28972, 1.0215)	
0.3	(0.0851, 0.7124)	(0.2256, 1.0038)	(0.2890, 1.0166)	
0.4	(0.0838, 0.7113)	(0.2279, 1.0108)	(0.2880, 1.0118)	
0.5	(0.0822, 0.7088)	(0.2305, 1.0178)	(0.2868, 1.0074)	
0.6	(0.0804, 0.7054)	(0.2325, 1.0247)	(0.2854, 1.0032)	
0.7	(0.0782, 0.7005)	(0.2368, 1.0313)	(0.2838, 0.9994)	
0.8	(0.0762, 0.6952)	(0.2403, 1.0375)	(0.2820, 0.9959)	
0.9	(0.0739, 0.6889)	(0.2438, 1.0429)	(0.2800, 0.9928)	
1	(0.0716, 0.6821)	(0.2473, 1.0476)	(0.27795, 0.9900)	
1.1	(0.0693, 0.6748)	(0.2507, 1.0515)	(0.2758, 0.9877)	
1.2	(0.0670, 0.6672)	(0.2540, 1.0545)	(0.2736, 0.9857)	
1.3	(0.0648, 0.6593)	(0.2570, 1.0567)	(0.2713, 0.9840)	
1.4	(0.0626, 0.6511)	(0.25985, 1.0581)	(0.2690, 0.9827)	
1.5	(0.0605, 0.6429)	(0.2624, 1.0588)	(0.2667, 0.9819)	
1.6	(0.0585, 0.6346)	(0.2647, 1.0589)	(0.2645, 0.9813)	
1.7	(0.0565, 0.6263)	(0.2668, 1.0584)	(0.2626, 0.9810)	
1.8	(0.0546, 0.6180)	(0.2686, 1.0575)	(0.2602, 0.9811)	
1.9	(0.0528, 0.6097)	(0.2701, 1.0561)	(0.2582, 0.9815)	
2	(0.0510, 0.6014)	(0.2714, 1.0545)	(0.2565, 0.9821)	
2.1	(0.0494, 0.5934)	(0.2726, 1.0527)	(0.2548, 0.9830)	
2.2	(0.0479, 0.5858)	(0.2735, 1.0507)	(0.2533, 0.9842)	
2.3	(0.0464, 0.5779)	(0.2742, 1.0486)	(0.2521, 0.9857)	
2.4	(0.0450, 0.5704)	(0.2748, 1.0464)	(0.2510, 0.9873)	
2.5	(0.0436, 0.5629)	(0.2752, 1.0442)	(0.2502, 0.9891)	
2.6	(0.0424, 0.5558)	(0.2754, 1.0420)	(0.2495, 0.9910)	
2.7	(0.0411, 0.5486)	(0.2756, 1.0399)	(0.2491, 0.9931)	
2.8	(0.0400, 0.5417)	(0.2756, 1.0377)	(0.2488, 0.9953)	
2.9	(0.0389, 0.5352)	(0.2755, 1.0357)	(0.2488, 0.9975)	
3	(0.0379, 0.5286)	(0.2754, 1.0338)	(0.2488, 0.9997)	
3.1	(0.0369, 0.5224)	(0.2752, 1.0319)	(0.2491, 1.0019)	
3.2	(0.0360, 0.5162)	(0.2749, 1.0301)	(0.2494, 1.0040)	(0.0064, 0.1806)
3.3	(0.0351, 0.5101)	(0.2746, 1.0285)	(0.2499, 1.0061)	(0.0154, 0.2819)
3.4	(0.0343, 0.5044)	(0.2742, 1.0269)	(0.2504, 1.0081)	(0.0227, 0.3444)
3.5	(0.0335, 0.4986)	(0.2737, 1.0255)	(0.2510, 1.0099)	(0.0289, 0.3901)
3.6	(0.0327, 0.4931)	(0.2732, 1.0242)	(0.2516, 1.0116)	(0.0340, 0.4253)
3.7	(0.0319, 0.4878)	(0.2728, 1.0230)	(0.2522, 1.0132)	(0.0383, 0.4530)
3.8	(0.0313, 0.4827)	(0.2723, 1.0218)	(0.2530, 1.0147)	(0.0418, 0.4755)
3.9	(0.0306, 0.4776)	(0.2718, 1.0208)	(0.2536, 1.0160)	(0.0448, 0.4936)
4	(0.0300, 0.4729)	(0.2713, 1.0199)	(0.2542, 1.0172)	(0.0470, 0.5080)
4.1	(0.0294, 0.4681)	(0.2708, 1.0191)	(0.2549, 1.0183)	(0.0490, 0.5205)
4.2	(0.0288, 0.4634)	(0.2703, 1.0184)	(0.2556, 1.0193)	(0.0505, 0.5303)
4.3	(0.0283, 0.4591)	(0.2698, 1.0177)	(0.2562, 1.0201)	(0.0518, 0.5387)
4.4	(0.0278, 0.4548)	(0.2693, 1.0172)	(0.2567, 1.0208)	(0.0528, 0.5456)
4.5	(0.0273, 0.4505)	(0.2688, 1.0166)	(0.2573, 1.0214)	(0.0536, 0.5516)
4.6	(0.0268, 0.4465)	(0.2684, 1.0162)	(0.2579, 1.0219)	(0.0542, 0.5560)
4.7	(0.0263, 0.4425)	(0.2680, 1.0159)	(0.2583, 1.0224)	(0.0547, 0.5601)
4.8	(0.0259, 0.4389)	(0.2676, 1.0156)	(0.2588, 1.0227)	(0.0550, 0.5632)
4.9	(0.0256, 0.4353)	(0.2672, 1.0153)	(0.2592, 1.0230)	(0.0550, 0.5632)
5	(0.0251, 0.4314)	(0.2668, 1.0151)	(0.2597, 1.0233)	(0.0553, 0.5675)
6	(0.0534, 0.5686)	(0.2642, 1.0149)	(0.2624, 1.0236)	(0.0222, 0.4019)
7	(0.0498, 0.5560)	(0.2634, 1.0224)	(0.2631, 1.0160)	(0.0207, 0.3823)
8	(0.0464, 0.5407)	(0.2637, 1.0213)	(0.2628, 1.0172)	(0.0204, 0.3732)
9	(0.0435, 0.5262)	(0.2637, 1.0205)	(0.2627, 1.0180)	(0.0218, 0.3777)
10	(0.0413, 0.5139)	(0.2636, 1.0200)	(0.2628, 1.0185)	(0.0257, 0.4021)

Table 2: Values of Hopf bifurcation points μ and ω for $\varepsilon = 0.15$, $\alpha = 0.55$, $d = 0.1$ and $p = 5$

μ as a function of τ . FM (FP) is for the equaion with the minus (plus) s

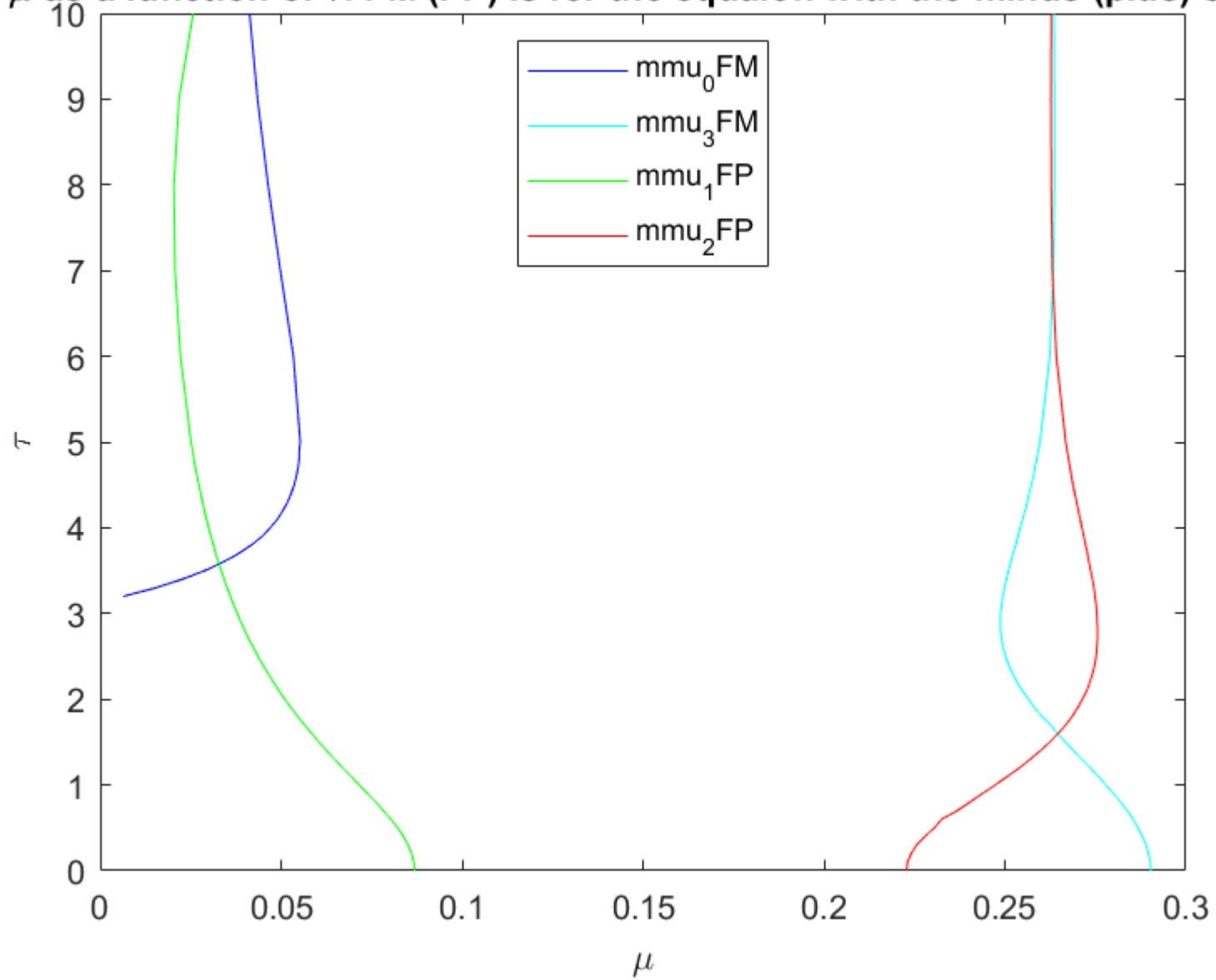


Figure 1: Caption

ω as a function of τ . FM (FP) is for the equaion with the minus (plus) s

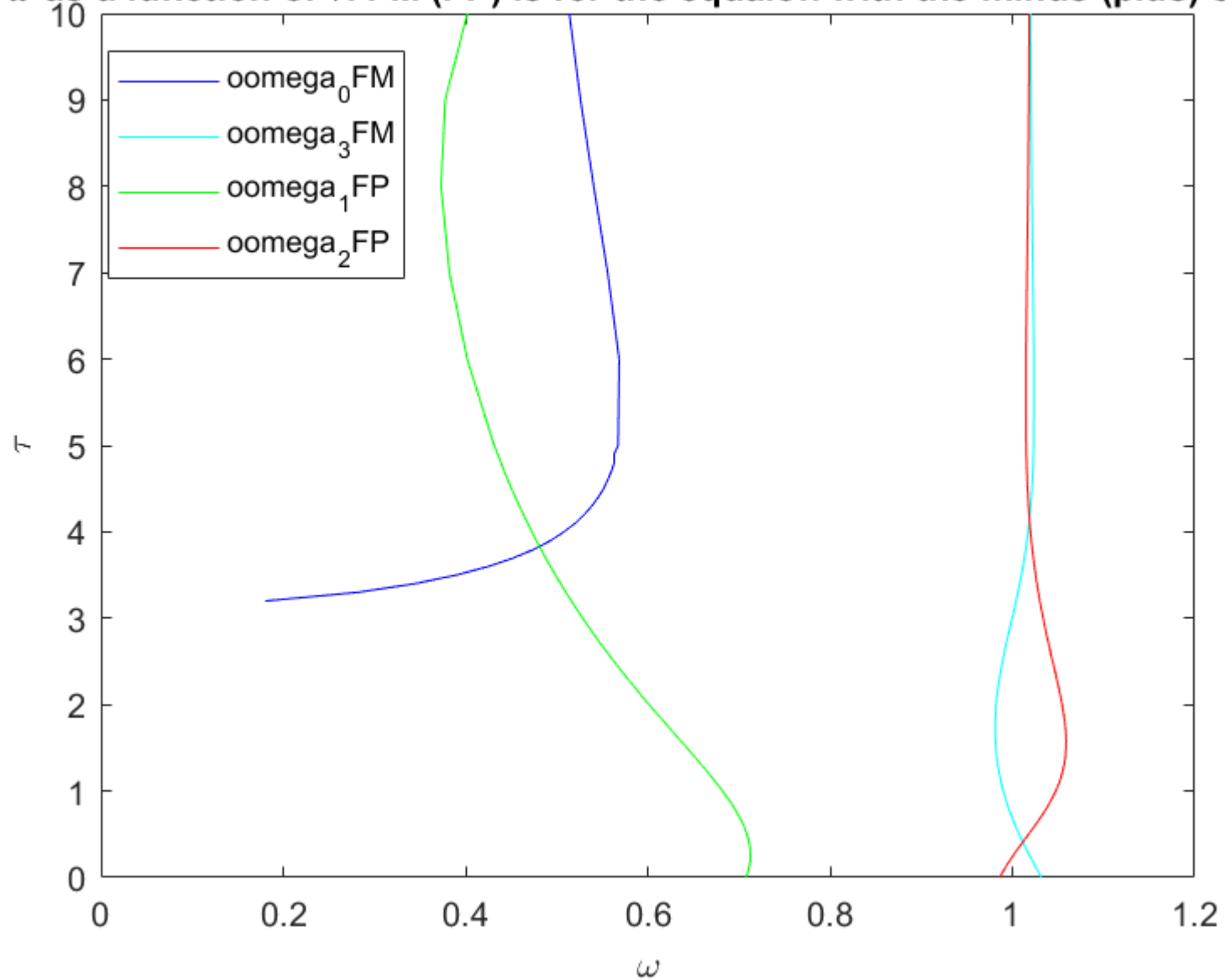


Figure 2: Caption